

VOSS DYNAMICS

Information, Representation, and Discovery

I. The Principle of Full Invertibility

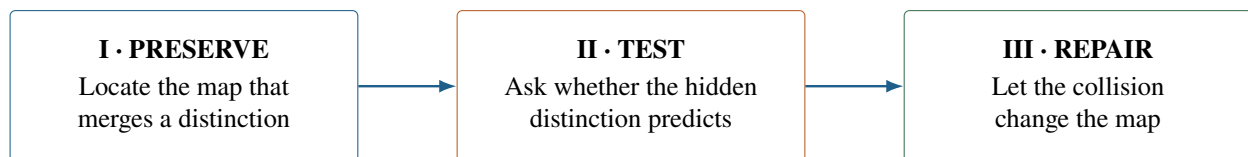
II. Emergent Predictive Representation

III. Pressure-Driven Invariant Synthesis

Logan Voss

One problem, three directions

The three studies address one problem from successive directions: how differences between possible states are preserved by dynamics, collapsed by representation, tested for predictive significance, and recovered when the current representation proves insufficient.



The trilogy begins by separating exact state, evolution, observation, numerical access, and boundary operations. It then asks whether distinctions hidden inside a representation fibre can change a future record. Finally, it makes the representation adaptive: a meaningful collision becomes a counterexample, a typed invariant program is synthesized, and the candidate is appended before a new audit.

The result is a single progression from provenance to predictive sufficiency to constructive repair. Each paper preserves its own assumptions, claim ledger, and evidence boundary.

Preserve the distinction. Test whether it matters. Repair the map when it fails.

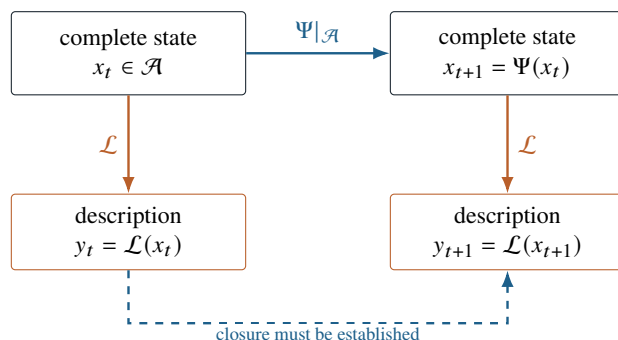
The Principle of Full Invertibility

Distinction provenance in dynamical systems and finite descriptions

Logan Voss

Voss Dynamics | Working mathematical thesis and computational audit

22 August 2026



Evidence statement. The inverse, determinant factorization, modular criterion, phase-contraction condition, closure criterion, and fibre-cardinality results are proved under their stated assumptions. Numerical experiments locate four distinct mechanisms of apparent loss: transition folding, many-to-one observation, finite-precision access, and explicit boundary clipping. All numerical results are internal model computations; no empirical physical measurement is reported. The paper develops a mathematical audit in which each mechanism receives its own type and provenance.

Abstract

The statement that information has been lost is incomplete until the relevant state, transition, description, inverse branch, and numerical representation are specified. This paper develops *distinction provenance*: a calculus for locating the operation at which previously distinct possibilities become identical or inaccessible. A model instantiates the Principle of Full Invertibility on an admissible domain $\mathcal{A}$ when its advance $\Psi|_{\mathcal{A}}$ is bijective while a separate description map $\mathcal{L} : \mathcal{A} \rightarrow \mathcal{Y}$ has nontrivial fibres.

For a drift–kick system with velocity-independent force, nonzero multiplier γ , and coupled phases, the translational inverse is algebraic and the full smooth-stratum Jacobian factors as

$$\det D\Psi = \gamma^{2N} \det D_{\phi} G_{\mathbf{r}^+}.$$

The force Jacobian cancels from the determinant. A configuration-dependent contraction bound certifies unique phase inversion, and a finite modular analogue is bijective exactly when its velocity multiplier is a unit. An explicit three-preimage target demonstrates the complementary global phenomenon: locally nonsingular phase branches can still fold onto one future.

The same framework separates description and access from transition dynamics. In the finite phrase census, 160 complete states produce 103 descriptions, with every checked shared-description pair remaining distinct. A forensic fibre theorem characterizes when retained evidence identifies one history. Fixed-partition entropy and float64 round-trip experiments show that observational uncertainty and numerical access evolve independently of exact-state identity. Together these results turn “where did the information go?” into a finite audit of the core transition, boundary operation, observation map, and numerical record.

Primary result Distinguishability has provenance. Every information-loss claim should identify the complete state, the map that advanced it, the map that described it, the branch structure of the inverse, and the numerical channel through which it was retained.

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1 Question, principle, and standard of proof

1.1 The question in two maps

Let $\mathcal{X}$ be a declared complete state space. Two maps play different roles:

$$\Psi : \mathcal{X} \rightarrow \mathcal{X}, \quad \mathcal{L} : \mathcal{X} \rightarrow \mathcal{Y}. \quad (1)$$

Ψ advances the modeled state. $\mathcal{L}$ records, names, bins, or summarizes it. The central discipline of this paper is to keep these maps separate. A collision under $\mathcal{L}$ is an observational identification; a collision under Ψ is a transition identification. Neither fact determines the other.

In this study the representation map $\mathcal{L}$ is fixed. Whether a distinction inside one fibre changes future predictions, and how the representation should be refined when it does, are separate downstream questions addressed by the later studies in the Voss Dynamics sequence.

Definition 1.1 (Complete modeled state). A state $x \in \mathcal{X}$ is *complete relative to the declared model* when it contains every coordinate required to evaluate one advance, while all remaining masses, frequencies, anchors, timestep choices, and control parameters are fixed and known.

Definition 1.2 (Distinction and provenance). A distinction is an ordered pair $x \neq x'$. Its *provenance* is the sequence of maps through which the pair is followed. A map preserves the distinction when its two images remain unequal.

Definition 1.3 (Description fibre). For $y \in \mathcal{Y}$, the fibre

$$\mathcal{L}^{-1}(y) := \{x \in \mathcal{X} : \mathcal{L}(x) = y\} \quad (2)$$

is the set of complete states represented by the same description.

Principle 1.4 (Full invertibility). On an admissible domain $\mathcal{A} \subseteq \mathcal{X}$, a model instantiates the Principle of Full Invertibility when

1. $\Psi|_{\mathcal{A}}$ is bijective onto $\mathcal{A}$;
2. $\mathcal{L}|_{\mathcal{A}}$ is a separate readout and may have nontrivial fibres;
3. every claim of loss identifies whether the responsible operation is the core advance, a boundary or controller, the readout, or the finite representation.

The word *full* refers to the declared state, not to universal applicability. The principle is a testable property of a specified map on a specified domain. This makes it useful even when a candidate domain contains both injective and folded regions: the same analysis locates the boundary.

1.2 Distinction provenance

Four map locations account for the cases studied here:

$\Psi_0(x) = \Psi_0(x'), x \neq x'$	core transition / inverse branches,	(3)
$\mathcal{B}(u) = \mathcal{B}(u'), u \neq u'$	boundary or controller collision,	
$\mathcal{L}(x) = \mathcal{L}(x'), x \neq x'$	observation collision,	
$\mathcal{N}(r) = \mathcal{N}(r'), r \neq r'$	finite-record collision or access limit.	

Branch multiplicity is the inverse signature of noninjectivity in the core transition, not a separate loss mechanism. The taxonomy converts an undifferentiated statement—“information disappeared”—into four map-specific questions.

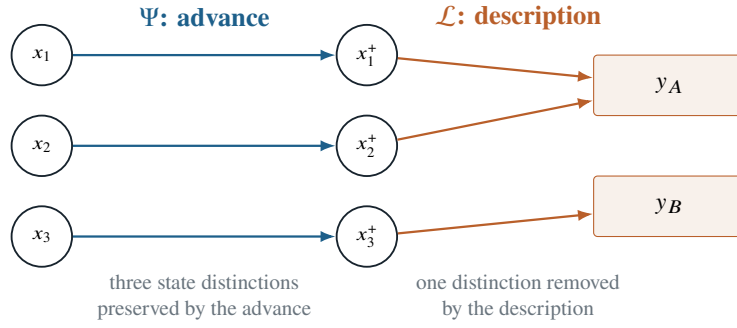


Figure 1: Two maps, two kinds of identity. The advance and the description are audited independently.

1.3 Claim discipline

Three statuses are used throughout:

PROVED

A mathematical consequence of explicitly stated assumptions.

COMPUTED

An exact symbolic, enumerative, deterministic, or seeded numerical result reproduced by code, with finite-precision protocols and sample counts declared where applicable.

OPEN

A theorem, domain characterization, or physical interpretation requiring further work.

1.4 Contribution ledger

Table 1: The paper’s central objects and present status.

Object	Result	Status
Drift-kick inverse	Algebraic recovery of velocity and position once a phase preimage is fixed.	PROVED
Full Jacobian	$\det D\Psi = \gamma^{2N} \det D_\phi G_{r^+}$ on smooth strata.	PROVED
Phase uniqueness	Configuration-dependent contraction certificate and exact two-clock fold criterion.	PROVED
Finite analogue	Permutation exactly when the modular multiplier is a unit.	PROVED
Description fibres	160 complete states map to 103 phrases; every checked shared-phrase pair is distinct.	COMPUTED
Inverse access	Float64 round-trip recovery quantified over contraction-certified trajectories.	COMPUTED
Global admissible domain	Characterization of a nontrivial invariant domain for the full phase-coupled map.	OPEN

The remainder of the paper develops these objects in the order required by the audit: state and advance, inverse structure, description fibres, finite access, and applications.

2 A structured dynamical realization

2.1 State manifold

The motivating realization contains N moving tokens and M fixed anchors. Token i carries

$$z_i = (\mathbf{r}_i, \mathbf{v}_i, \theta_i, \phi_i), \quad \mathbf{r}_i, \mathbf{v}_i \in \mathbb{R}^2, \quad \theta_i, \phi_i \in \mathbb{T} := \mathbb{R}/(2\pi\mathbb{Z}). \quad (4)$$

Here $\mathbf{r}_i$ and $\mathbf{v}_i$ are position and velocity, θ_i is a directional bias, and ϕ_i is an internal phase. Masses m_i , intrinsic frequencies ω_i , timestep Δt , anchor data, and force parameters are fixed and known. The dynamical state is

$$x = (\mathbf{r}, \mathbf{v}, \boldsymbol{\theta}, \boldsymbol{\phi}) \in \mathcal{X}_N := (\mathbb{R}^2 \times \mathbb{R}^2 \times \mathbb{T} \times \mathbb{T})^N. \quad (5)$$

Each anchor a has position $\mathbf{p}_a$, positive weight Q_a , and preferred angles (θ_a, ϕ_a) . Anchors are parameters of the one-step map. Any protocol that changes them defines an enlarged or externally controlled system and is treated separately.

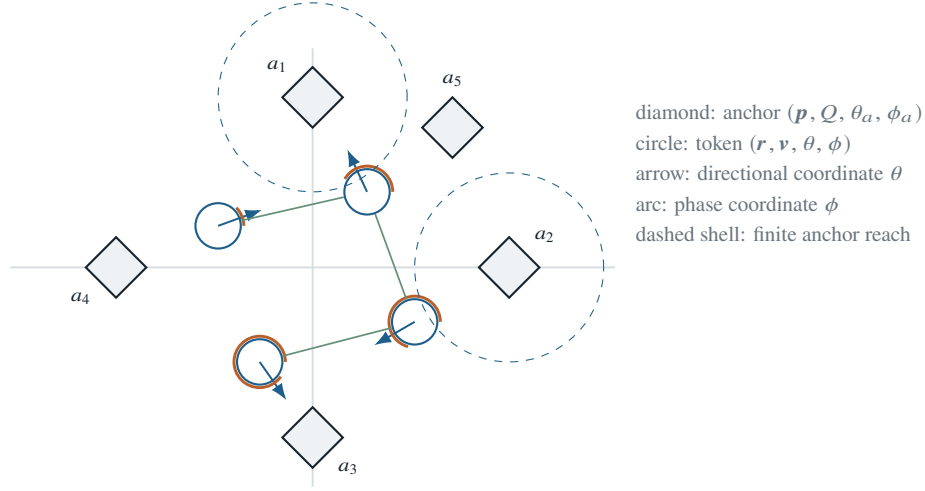


Figure 2: A structured field realization. Translational, directional, and phase coordinates coexist in one declared state.

2.2 Ordered advance

The practical discrete realization is a drift, a force kick, and a phase update:

$$\mathbf{r}_i^+ = \mathbf{r}_i + \mathbf{v}_i \Delta t, \quad (6a)$$

$$\mathbf{v}_i^+ = \gamma \mathbf{v}_i + \frac{\Delta t}{m_i} \mathbf{F}_i(\mathbf{r}^+, \boldsymbol{\theta}, \boldsymbol{\phi}), \quad (6b)$$

$$\phi_i^+ = \left[\phi_i + \omega_i \Delta t + \beta \Delta t \sum_{j \neq i} \frac{\sin(\phi_j - \phi_i)}{\|\mathbf{r}_i^+ - \mathbf{r}_j^+\| + \varepsilon} \right]_{2\pi}, \quad (6c)$$

$$\theta_i^+ = \theta_i. \quad (6d)$$

The structural fact is the update order. The force is evaluated at the advanced position and old angles, and it does not depend on velocity. This triangular ordering controls both the inverse and the determinant.

The force inventory is

$$\mathbf{F}_i = \mathbf{F}_i^{\text{anchor}} + \mathbf{F}_i^{\text{wave}} + \mathbf{F}_i^{\text{pair}} + \mathbf{F}_i^{\text{control}}. \quad (7)$$

The first term supplies finite-range directional bias, the second is a softened phase-bearing radial field, and the third combines short repulsion with intermediate attraction. A fixed external control field can be included without changing the translational inverse. The exact implementation is recorded in [appendix B](#); the inverse theorems require only velocity independence and fixed known parameters.

2.3 Phase map

At a known advanced position $\mathbf{r}^+$, define

$$G_{\mathbf{r}^+}(\boldsymbol{\phi}) := [\boldsymbol{\phi} + \omega \Delta t + \beta \Delta t \mathbf{K}(\boldsymbol{\phi}; \mathbf{r}^+)]_{2\pi}, \quad (8)$$

where

$$K_i(\boldsymbol{\phi}; \mathbf{r}^+) = \sum_{j \neq i} \frac{\sin(\phi_j - \phi_i)}{\|\mathbf{r}_i^+ - \mathbf{r}_j^+\| + \varepsilon}. \quad (9)$$

The full advance is therefore the composition of an explicitly invertible mechanical block and an implicit phase block. This decomposition makes the source of every inverse branch visible.

2.4 Core and variants

Three operations are kept distinct:

Core advance: [equation \(6\)](#) with no clipping and fixed parameters.

Boundary variant: an added wall rule, whose injectivity depends on how overshoot is represented.

Feedback variant: a description-dependent controller, which becomes part of the complete state if it affects future dynamics.

The main theorems concern the core advance. Boundary and feedback operations are valuable because they show how additional maps alter distinction provenance.

3 Exact inverse structure

3.1 Triangular inverse

Theorem 3.1 (Conditional exact inverse). *Fix a wall-free one-step advance [equation \(6\)](#). If $\gamma \neq 0$ and the phase map G_{r^+} is bijective at the advanced position, then the complete pre-state is uniquely recovered by*

$$\phi = G_{r^+}^{-1}(\phi^+), \quad (10a)$$

$$v = \gamma^{-1} [v^+ - \Delta t M^{-1} F(r^+, \theta, \phi)], \quad (10b)$$

$$r = r^+ - v \Delta t, \quad (10c)$$

$$\theta = \theta^+. \quad (10d)$$

Proof. The final position is known. Bijectivity of G_{r^+} determines the old phase. The force is then known because it depends on (r^+, θ, ϕ) and fixed parameters, but not on old velocity. Since $\gamma \neq 0$, [equation \(10b\)](#) determines velocity; the drift then determines position. Direct substitution verifies both compositions. $\square$

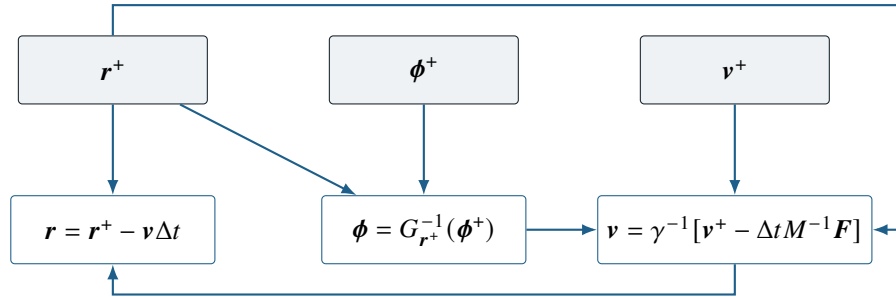


Figure 3: The inverse is triangular: phase branch first, then force and velocity, then position.

The mechanical sector is therefore globally algebraic for every velocity-independent force when $\gamma \neq 0$. All branch structure enters through the phase map.

3.2 Full determinant factorization

Theorem 3.2 (Full Jacobian). *At every smooth point of the wall-free map with static θ ,*

$$\boxed{\det D\Psi = \gamma^{2N} \det D_\phi G_{r^+}(\phi)}. \quad (11)$$

The determinant is independent of positional and phase derivatives of the force.

Proof. First change coordinates from (r, v, ϕ) to $(r^+ = r + \Delta t v, v, \phi)$; this block-triangular map has determinant one. In these coordinates, the remaining Jacobian is

$$\begin{pmatrix} I & 0 & 0 \\ * & \gamma I & \Delta t M^{-1} D_\phi F \\ * & 0 & D_\phi G_{r^+} \end{pmatrix}.$$

Multiplying the diagonal block determinants yields [equation \(11\)](#). $\square$

Corollary 3.3 (Local singularity channels). *The smooth full map is locally invertible exactly where*

$$\gamma \neq 0 \quad \text{and} \quad \det D_\phi G_{r^+} \neq 0.$$

Force curvature cannot create a local singularity in this update ordering.

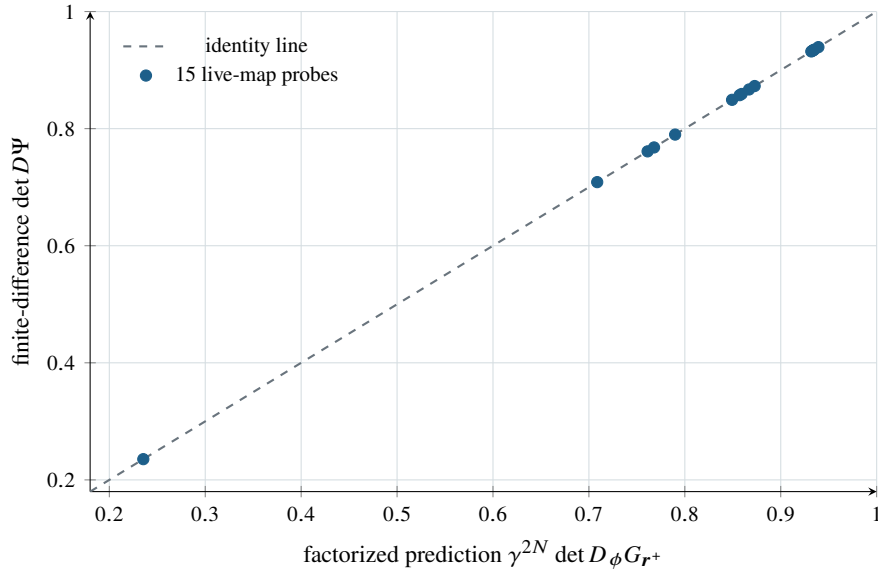


Figure 4: Numerical audit of [equation \(11\)](#). Maximum relative discrepancy is 2.1×10^{-8} .

Source: evidence/thesis_verification.json, determinant_factorization.

3.3 Phase uniqueness

The phase Jacobian is

$$\begin{aligned} (D_\phi G)_{ii} &= 1 - \beta \Delta t \sum_{j \neq i} \frac{\cos(\phi_j - \phi_i)}{\tilde{d}_{ij}^+}, \\ (D_\phi G)_{ij} &= \beta \Delta t \frac{\cos(\phi_j - \phi_i)}{\tilde{d}_{ij}^+}, \quad i \neq j, \end{aligned} \tag{12}$$

where $\tilde{d}_{ij}^+ = \|\mathbf{r}_i^+ - \mathbf{r}_j^+\| + \varepsilon$.

Theorem 3.4 (Configuration-dependent contraction certificate). *Define*

$$q(\mathbf{r}^+) := 2 |\beta| \Delta t \max_i \sum_{j \neq i} \frac{1}{\tilde{d}_{ij}^+}. \tag{13}$$

If $q(\mathbf{r}^+) < 1$, then the inverse phase iteration is a contraction in the ℓ_∞ norm on a consistent torus lift. The phase preimage is unique in that lift, fixed-point inversion converges geometrically, and G_{r^+} has lower bi-Lipschitz bound $1 - q$.

The certificate is configuration-dependent. It supplies an operational gate for every trajectory used in the round-trip study.

3.4 Global branch geometry

For two tokens at fixed softened separation D , let

$$\delta = \phi_1 - \phi_2, \quad \Delta\omega = \omega_1 - \omega_2, \quad a = \frac{2\beta\Delta t}{D}.$$

The exact relative-phase map is

$$\boxed{\delta^+ = \delta + \Delta\omega \Delta t - a \sin \delta \pmod{2\pi}.} \tag{14}$$

For equal frequencies, its derivative is $1 - a \cos \delta$. The circle map is a diffeomorphism for $|a| < 1$, a singular monotone homeomorphism at $|a| = 1$, and noninjective for $|a| > 1$.

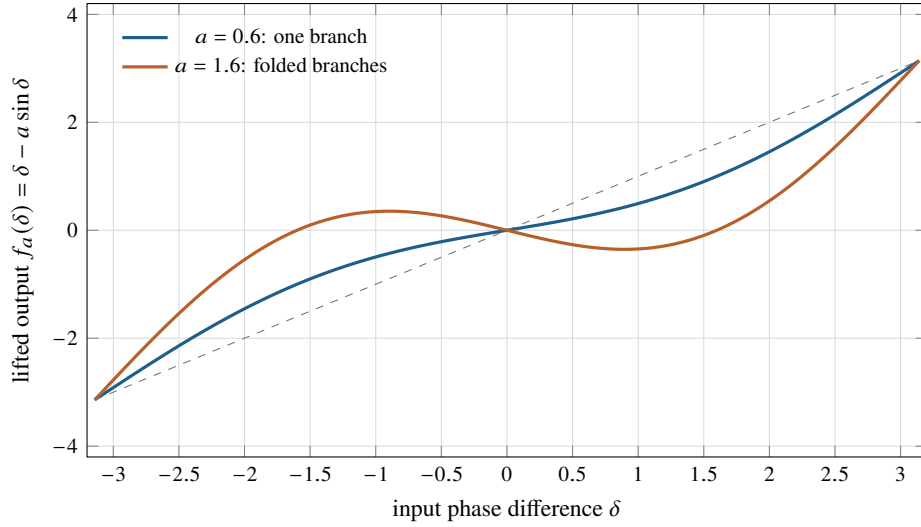


Figure 5: The same phase law contains an injective regime and a folded regime. Distinction provenance locates the transition between them.

The live eight-token system reaches the folded regime at default parameters. At seed 22, step 9, one target has $q = 1.338984$ and three refined phase roots. Combining each root with the algebraic mechanical inverse produces three complete pre-states.

Table 2: Three complete pre-states for one target. Every phase Jacobian is locally nonsingular; the distinction is global branch structure.

root	$\det D_\phi G$	distance from generating pre-state	forward residual
0	-0.2300	1.09×10^{-15}	0
1	0.4584	7.386	1.50×10^{-13}
2	0.4115	6.738	1.33×10^{-13}

Source: evidence/thesis_verification.json, phase_multiroot_collision.

This example demonstrates the value of the framework: a nonzero local Jacobian, a convergent inverse solver, and global branch uniqueness are separate questions.

3.5 Finite analogue

On $\mathbb{Z}/M\mathbb{Z}$, consider

$$r^+ = r + v \pmod{M}, \quad v^+ = g v + F(r^+) \pmod{M}. \quad (15)$$

Theorem 3.5 (Modular unit criterion). *Equation (15) is a permutation of the M^2 states for every force F if and only if $\gcd(g, M) = 1$.*

Proof. If g is a unit,

$$v = g^{-1}[v^+ - F(r^+)] \pmod{M}, \quad r = r^+ - v \pmod{M}.$$

If g is not a unit, multiplication by g has a nontrivial kernel, and compensated position–velocity pairs collide. $\square$

Enumeration gives 63,001/63,001 images for $(M, g) = (251, 3)$, while $(M, g) = (16, 2)$ yields 128/256 images. The theorem provides a complete finite model of the distinction between invertible multiplication and state aggregation.

Remark 3.6 (Six independent properties). Bijectivity, local nonsingularity, time-reversal symmetry, volume preservation, energy conservation, and stable numerical inversion are distinct properties. The results above identify exactly which of them follow from each assumption.

4 Description fibres and predictive closure

4.1 A finite phrase readout

The implemented readout groups tokens by nearby anchors or marks them FREE. Let $C(x)$ be this deterministic clustering and let $P(C(x))$ be the resulting phrase. The observation map is

$$\mathcal{L}(x) := P(C(x)). \quad (16)$$

It retains cluster labels and counts while omitting velocities, directional coordinates, phases, exact positions, and within-cluster identity. An auxiliary geometric score,

$$S_{\text{geom}}(x) = S_{\text{name}}(C(x)) + S_{\text{resid}}(x \mid C(x)), \quad (17)$$

summarizes phrase complexity and positional residual. The score is not needed for the fibre result; the many-to-one structure follows directly from [equation \(16\)](#).

Across 160 independently evolved 14-token states,

$$160 \text{ complete states} \xrightarrow{\mathcal{L}} 103 \text{ phrases}. \quad (18)$$

Thirty-one phrases occur more than once. Every one of 68 checked shared-phrase pairs is a pair of distinct complete states, and the largest computed fibre contains eight states.

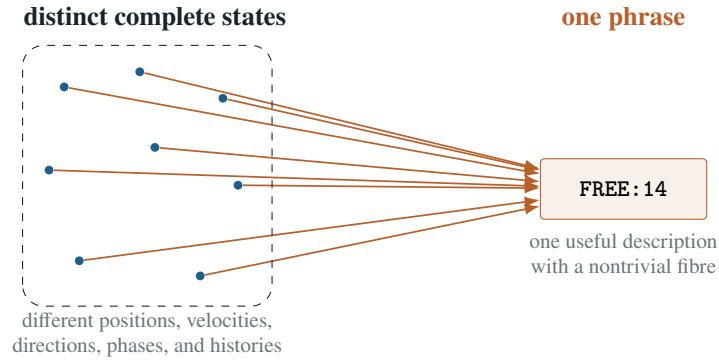


Figure 6: A computed observation fibre containing eight distinct complete states.

Source: results/battery.json, mdl_many_to_one.

4.2 History fibres

Let $y = \mathcal{L}(x_t)$ be retained final evidence and suppose $\Psi|_{\mathcal{A}}$ is bijective. The k -step histories compatible with y are

$$\mathcal{H}_k(y) := \{\Psi^{-k}(x) : x \in \mathcal{L}^{-1}(y) \cap \mathcal{A}\}. \quad (19)$$

Theorem 4.1 (Forensic fibre theorem). *If $\Psi|_{\mathcal{A}}$ is bijective, then*

$$|\mathcal{H}_k(y)| = |\mathcal{L}^{-1}(y) \cap \mathcal{A}|. \quad (20)$$

Consequently, retained evidence identifies one k -step history exactly when its reachable observation fibre contains one complete state.

Proof. The restriction of Ψ^{-k} to any subset of $\mathcal{A}$ is a bijection onto its image and therefore preserves cardinality. $\square$

The theorem separates a unique underlying predecessor from a uniquely inferable predecessor. Inverse dynamics transports the ambiguity already present in the evidence; it does not remove it.

4.3 Description-level closure

A many-to-one readout does not automatically define an autonomous macro-dynamics. The additional requirement is fibre compatibility, the set-level form of exact lumpability or semiconjugacy [1, 2].

Proposition 4.2 (Closure criterion). *Let $\mathcal{L} : \mathcal{X} \rightarrow \mathcal{Y}$ be surjective. A deterministic map $T : \mathcal{Y} \rightarrow \mathcal{Y}$ satisfying*

$$T \circ \mathcal{L} = \mathcal{L} \circ \Psi \quad (21)$$

exists if and only if

$$\mathcal{L}(x) = \mathcal{L}(x') \implies \mathcal{L}(\Psi x) = \mathcal{L}(\Psi x'). \quad (22)$$

Proof. If T exists, equation (22) follows immediately. Conversely, define $T(y) = \mathcal{L}(\Psi x)$ for any $x \in \mathcal{L}^{-1}(y)$; fibre compatibility makes the value independent of the chosen representative. $\square$

If fibre compatibility holds for both Ψ and Ψ^{-1} , the induced macro-map can itself be bijective. If it fails, the reduced process requires hidden coordinates, memory, or stochastic closure, as in projection-operator methods [3–5]. This is the precise bridge from description fibres to effective dynamics.

4.4 Records and temporal order

Complete-state retrodictability and record order are separate structures. A bijective transition pairs adjacent complete states in both directions. A recorder, by contrast, selects and stores a sequence

$$y_t = \mathcal{L}(x_t), \quad y_{t+1} = \mathcal{L}(x_{t+1}), \dots$$

whose fibres may be nontrivial. Directional records can therefore coexist with distinction-preserving state evolution. Whether those records support a Markovian, reversible, or history-dependent macro-law is decided by proposition 4.2, not by the cardinality of a single fibre alone.

5 Information and finite access

5.1 Distinctions, probability, and volume

For a finite set $S \subset \mathcal{X}$, define its distinction information by

$$I_{\text{dist}}(S) := \log_2 |S|.$$

Every injective map preserves I_{dist} on finite subsets. This is the invariant used by the principle: distinct complete alternatives remain distinct.

Theorem 5.1 (Discrete entropy under a bijection). *If X is a finite random state and $Y = f(X)$ for a bijection f , then*

$$H(Y) = H(X)$$

for every probability distribution on X .

Proof. The output probabilities are a permutation of the input probabilities. $\square$

The finite modular experiment verifies this identity for a nonuniform distribution to 3.6×10^{-15} bits. Under the $g = 0$ many-to-one map, a uniform M^2 -state distribution becomes an M -image distribution and loses exactly $\log_2 M$ bits.

Continuous differential entropy obeys a different law. Let X have an absolutely continuous distribution, let Ψ be a C^1 diffeomorphism on its support, and assume $h(X)$ and $\mathbb{E} |\log |\det D\Psi(X)||$ are finite. Then

$$h(\Psi(X)) = h(X) + \mathbb{E} \log |\det D\Psi(X)|. \quad (23)$$

Thus a bijection may contract volume and change differential entropy while preserving every exact state distinction. Invertible dissipative maps provide established examples of this separation [6–9].

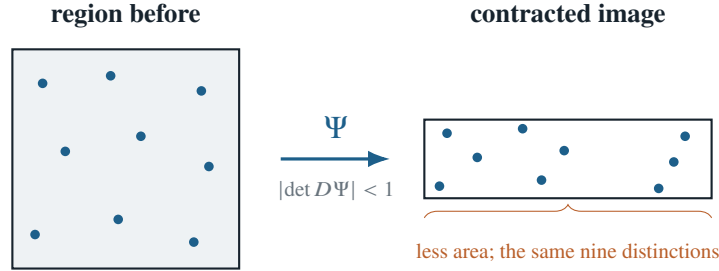


Figure 7: Volume contraction and state identification answer different questions.

5.2 A fixed observational partition

For a quantizer $Q : \mathcal{X} \rightarrow \mathcal{B}$, the coarse entropy $H(Q(X_t))$ depends on the chosen partition. The corrected ensemble diagnostic declares fixed position, velocity, and circular phase bins before evolution, retains overflow bins, and reports the sum of one-dimensional marginal entropies.

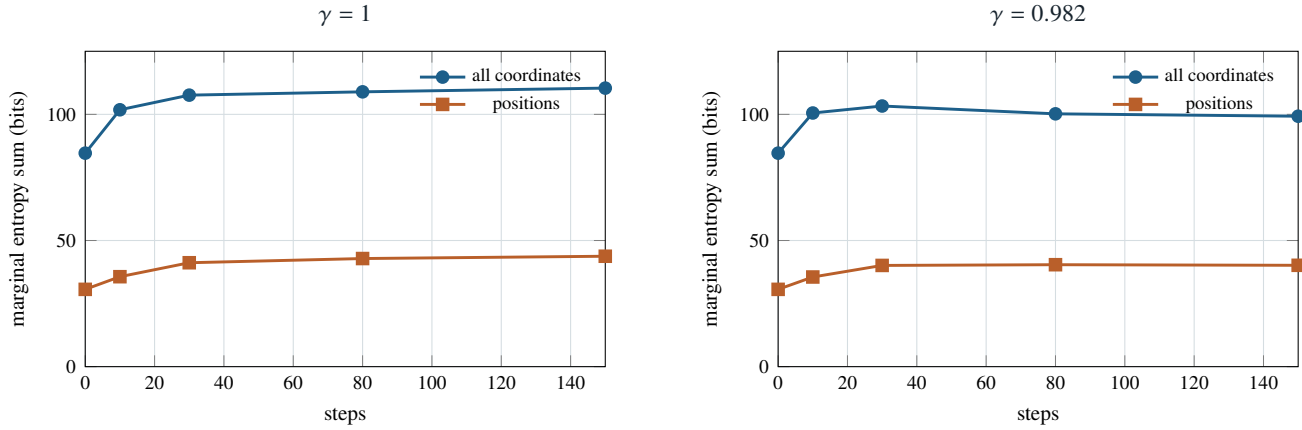


Figure 8: A fixed observer partition evolves differently from exact distinction cardinality. The plotted quantity is a sum of marginals, not a joint state entropy.

Source: evidence/thesis_verification.json, fixed_partition_marginal_entropy.

The $\gamma = 1$ full marginal sum rises from 84.62 to 110.40 bits by $t = 150$. For $\gamma = 0.982$, it rises to 103.31 bits at $t = 30$ and settles at 99.26 bits by $t = 150$. These values describe the observer’s fixed alphabet. They do not count exact-state collisions.

5.3 A state is not a snapshot

Projection onto position,

$$\pi_r(x) = r,$$

removes velocity and phase coordinates required by the inverse. In a 10-token, 30-step probe, the complete final state returns to its origin with $L^2 = 1.31 \times 10^{-10}$. Zeroing final velocities first gives $L^2 = 15.35$; zeroing phases gives 25.37. The scale separation directly demonstrates state sufficiency: a visible configuration and a complete dynamical state are different objects.

5.4 Float64 round-trip access

On a trajectory for which every phase update satisfies $q < 1$, the exact branch is unique. Finite arithmetic introduces a second question: how accurately can that branch be recovered from the stored endpoint? Linearization gives the schematic amplification

$$\|\delta x_{t-T}\| \lesssim \left(\prod_{s=0}^{T-1} \|D\Psi^{-1}(x_{t-s})\| \right) \|\epsilon_t\| + \text{accumulated roundoff}. \quad (24)$$

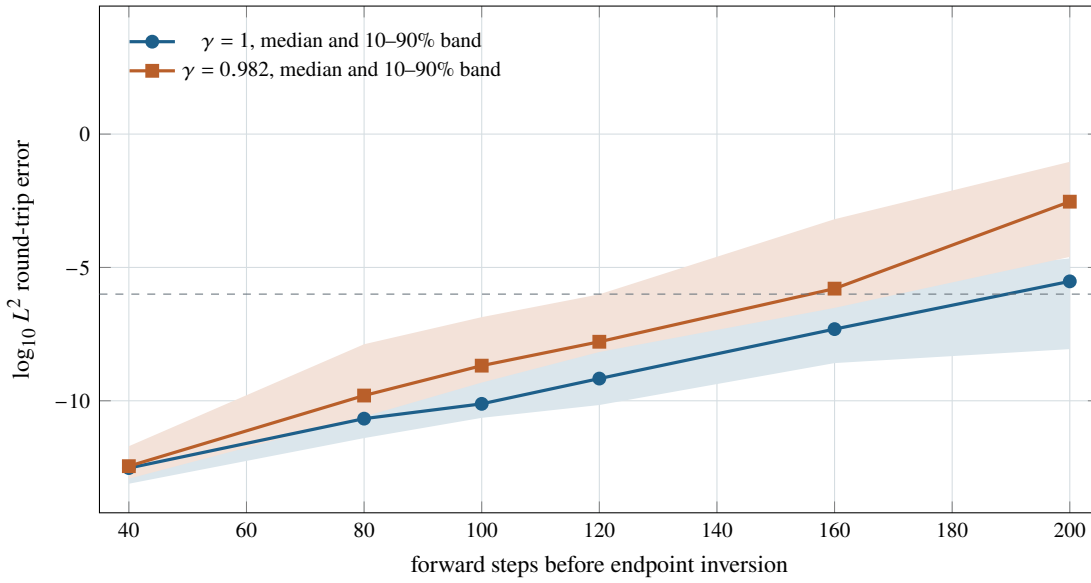


Figure 9: Float64 round-trip recovery on trajectories certified by $q < 1$ and a converged wrapped phase residual.

Source: evidence/thesis_verification.json; 144 endpoint inversions before certification filters.

At $T = 200$, the certified median error is 3.02×10^{-6} for $\gamma = 1$ over 12 seeds and 3.04×10^{-3} for $\gamma = 0.982$ over 8 seeds. The damped 10–90% range spans 2.86×10^{-5} to 1.69. This trajectory- and precision-dependent access scale is an *epistemic horizon*: the branch may remain mathematically unique after the stored endpoint ceases to recover it at the desired tolerance.

6 Distinction provenance in action

6.1 Four located mechanisms

The model and evidence package exhibit four distinct outcomes under one common audit.

Table 3: Distinction provenance across the audited model.

Mechanism	Diagnostic object	Located result
Transition folding	Phase preimages of one target	Three complete pre-states share one future; the loss occurs in the phase advance.
Boundary clipping	Two in-domain overshoots	States at distance 0.2 become identical under the clamp; the loss occurs at the wall operation.
Observation fibre	Anchor-labelled phrase	160 states produce 103 phrases; the loss occurs in the readout.
Finite access	Endpoint inverse in float64	Recovery error grows with horizon on unique certified branches; the loss is practical access at a chosen tolerance.

These outcomes are not competing explanations of the same event. The core transition and an optional boundary or controller act in sequence; description and numerical storage are separate representations of the retained complete state:

$$\begin{array}{ll}
 x \xrightarrow{\Psi_0} u \xrightarrow{\mathcal{B}} x^+, & \text{core advance and optional boundary,} \\
 x^+ \xrightarrow{\mathcal{L}} y, & \text{description branch,} \\
 x^+ \xrightarrow{\mathcal{N}} \hat{x}^+, & \text{stored numerical-state branch.}
 \end{array} \tag{25}$$

Here $\mathcal{B}$ may be the identity when no boundary or controller is present. The provenance question asks for the first map at which a chosen distinction is merged or becomes inaccessible.

6.2 Boundary clipping as an exact operation

Proposition 6.1 (Wall-clamp collision). *Let one force-free coordinate have wall w , equal velocities $v > 0$, and two in-domain positions $r_1 \neq r_2 < w$ satisfying*

$$r_1 + v\Delta t > w, \quad r_2 + v\Delta t > w.$$

If the boundary rule replaces every overshoot by w and multiplies both velocities by the same bounce factor, the two states have one image.

Proof. The clamp sends both advanced positions to w ; equal velocities receive the same update; all other coordinates are equal. $\square$

The computational construction uses $w = 3.6$, pre-state distance 0.2, and produces final distance zero. The proposition gives a precise engineering lesson: an invertible boundary must retain overshoot, for example by periodic wrapping on a torus or exact specular reflection.

6.3 Relational phase transport

The two-clock law [equation \(14\)](#) also yields an exact retained relation. When coupling is switched off after step n_0 ,

$$\delta_n = \delta_{n_0} + (n - n_0)\Delta\omega \Delta t \pmod{2\pi}. \quad (26)$$

Equivalently, the co-rotating coordinate

$$I_n := \delta_n - n\Delta\omega \Delta t$$

is conserved. Equal frequencies reduce this to conservation of the raw phase difference.

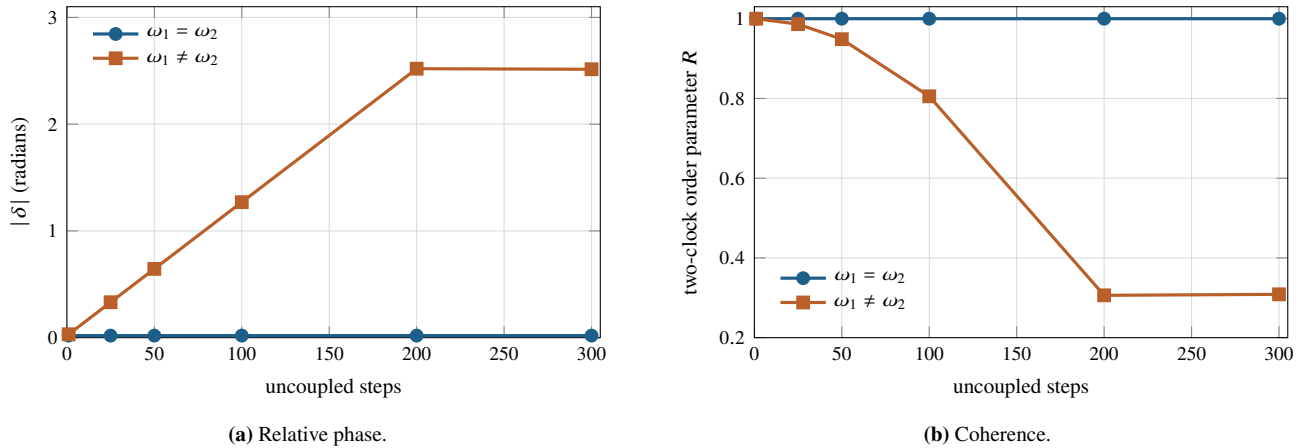


Figure 10: After four coupling steps, β is set exactly to zero. Equal-frequency phase difference is retained; unequal-frequency drift follows [equation \(26\)](#) to floating-point precision.

Source: evidence/thesis_verification.json, decoupled_clock_memory_and_readout.

6.4 Controlled local response

Resetting one phase is an intervention. Clone one phase state into a reset arm and a no-reset control, alter only the separation, and compare the partner after one step. The exact difference is

$$\Delta_{\text{reset}}\phi_2^+ = \frac{\beta\Delta t}{\tilde{D}} \left[\sin(\phi_{1,\text{reset}} - \phi_2) - \sin(\phi_{1,\text{control}} - \phi_2) \right]. \quad (27)$$

Measured effects are 0.32266 radians at distance 0.3 and 0.004047 radians at distance 24, matching the softened $1/\tilde{D}$ ratio. The partner does not change in the same instant; the next advance transmits the controlled response. This supplies a clean example of relational memory, local intervention, and distance-weighted dynamics within one exact phase law.

6.5 What the demonstrations establish

The multiroot, wall, phrase, and round-trip experiments form a single positive result: the framework identifies *which operation* owns each loss of distinguishability. This is stronger than labeling every ambiguity “entropy” or every failed reconstruction “irreversibility.” It produces a typed result that can be carried into other dynamical, observational, and computational systems.

7 Implications of distinction provenance

7.1 A five-question audit

The framework reduces an information-loss claim to five questions:

1. What is the complete modeled state?
2. Which map advances it?
3. Which map describes or records it?
4. Is the relevant inverse branch unique?
5. Is that branch recoverable at the available precision?

The order matters. A partial record cannot answer a question about complete-state identity until the observation fibre is characterized; a numerical inverse cannot establish uniqueness until branch structure is known.

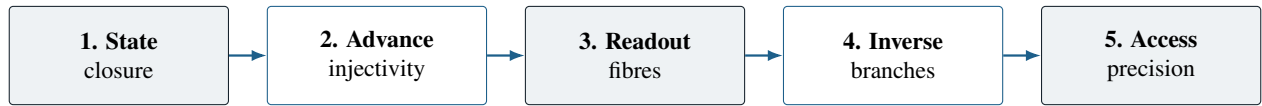


Figure 11: The distinction-provenance audit. Each stage answers a different identity question.

7.2 Dynamics and information theory

The determinant theorem separates local volume change from state identity. This is consistent with established invertible dissipative maps [6–8]: contraction is a geometric property of an ensemble, while injectivity is a property of point identity. The phase-fold example adds the complementary lesson that nonzero local Jacobians do not settle global branch uniqueness.

The observation map introduces a second, independent information structure. Discrete Shannon entropy is invariant under a finite permutation [10–12]; differential entropy changes by the expected log-Jacobian; coarse entropy depends on a partition; and description fibres determine which complete states a record can distinguish. Treating these as separate quantities gives entropy statements a precise mathematical owner.

7.3 Causality and records

Invertibility, time-reversal symmetry, intervention direction, thermodynamic irreversibility, and record order need not coincide. The transition map may admit a unique inverse without possessing a physical time-reversal involution. A recorder may accumulate descriptions in one order even when adjacent complete states determine one another. An autonomous macro-law exists only under the closure criterion [proposition 4.2](#).

This decomposition replaces a single overloaded word with explicit structures:

transition orientation, retrodictive relation, intervention response, record order.

Each can be tested without assigning the properties of one to another.

7.4 Forensic reconstruction

[Theorem 4.1](#) gives a direct rule for reconstruction. If the reachable observation fibre is a singleton, the evidence identifies one complete present and therefore one history on a bijective domain. If the fibre contains several states, the scientifically correct output is the transported history family $\mathcal{H}_k(y)$.

This distinction is immediately applicable to logs, simulations, market records, event traces, and physical measurements. It separates

one modeled past from one past identified by retained evidence.

The framework therefore supports both exact reconstruction and principled ambiguity reporting.

7.5 Simulation, telemetry, and digital twins

In software and scientific computing, the two-map architecture becomes

$$\text{state transition} \quad || \quad \text{telemetry summarization}.$$

A dashboard collision is not a state collision. Conversely, a dashboard cannot reconstruct state unless its fibre is known to be a singleton on the reachable set. Systems designed for later inversion should:

1. retain every closure variable and controller parameter;
2. preserve intervention and boundary histories;
3. checkpoint before round-trip error exceeds the required tolerance;
4. validate recovery over ensembles rather than one trajectory;
5. report solver convergence separately from branch uniqueness.

The float64 study turns these principles into an instrumentation objective: extend the recovery horizon by selecting state coordinates, precision, and checkpoint cadence deliberately.

7.6 Observation design

The fibre viewpoint gives every sensor system a measurable target. A readout is adequate for a task when its reachable fibres preserve the distinctions required by that task. This is weaker than demanding a globally injective sensor and stronger than assuming a low-error summary is sufficient.

Delay-coordinate methods provide one path from partial observation to state reconstruction [13, 14]. The relevant question is not whether a single snapshot is complete, but whether a declared history of observations embeds the reachable state set with acceptable conditioning.

7.7 A transferable scientific method

The resulting workflow is

$$\boxed{\text{declare} \longrightarrow \text{advance} \longrightarrow \text{observe} \longrightarrow \text{invert} \longrightarrow \text{locate}.} \quad (28)$$

Declare the complete state and parameters. Advance by a specified map. Observe through an explicit readout. Invert with branch and precision diagnostics. Locate the first operation that removes or obscures the distinction of interest. This is the principal methodological contribution of the paper.

8 Scope and open theorem program

8.1 Scope of the present results

The paper establishes a conditional principle and a model-level audit. Its scope is defined by the following boundaries:

1. The translational inverse is global for $\gamma \neq 0$ and velocity-independent force; the full inverse additionally requires a unique phase branch.
2. The certificate $q(r^+) < 1$ proves uniqueness for an individual phase update. A nontrivial invariant domain on which it holds for all time remains to be characterized.
3. Jacobian and differential-entropy identities apply on smooth strata. The finite anchor cutoffs are piecewise smooth.
4. The phrase readout is a deterministic geometric clustering with an auxiliary score. It is not required to be a complete code.
5. The round-trip experiment measures end-to-end float64 recovery, not an independently estimated condition number.
6. The model separates exact-state, observational, and numerical information. A thermodynamic theory would additionally require energy, reservoirs, temperature, and a physical measure.
7. The equations define a mathematical system. Applying the principle to a physical domain requires an independently justified complete state, calibrated parameters, and prospective tests.

These boundaries delimit the theorem without reducing its generality as an audit method.

8.2 Global phase domain

The phase Jacobian is an identity minus a signed weighted graph Laplacian. The principal open theorem is to characterize the largest domain on which G_r is globally bijective and invariant under the full advance. Promising routes include:

1. spectral bounds sharper than the ℓ_∞ certificate $q < 1$;

2. degree-theoretic continuation of the torus map;
3. explicit branch-bifurcation analysis beyond the two-clock reduction;
4. invariant minimum-separation estimates for the coupled translational dynamics.

8.3 Invertible boundaries

The clamp in [proposition 6.1](#) can be replaced by a distinction-preserving boundary:

- periodic position on a two-torus;
- exact specular reflection retaining overshoot;
- an enlarged state that records the boundary impulse and traversed distance.

Each option yields a different but explicitly auditable state space.

8.4 Continuous-time realization

A smooth continuous counterpart is

$$\dot{\mathbf{r}}_i = \mathbf{v}_i, \quad (29a)$$

$$\dot{\mathbf{v}}_i = -\kappa \mathbf{v}_i + m_i^{-1} \mathbf{F}_i(\mathbf{r}, \boldsymbol{\theta}, \boldsymbol{\phi}), \quad (29b)$$

$$\dot{\phi}_i = \omega_i + \beta \sum_{j \neq i} \frac{\sin(\phi_j - \phi_i)}{\|\mathbf{r}_i - \mathbf{r}_j\| + \varepsilon}. \quad (29c)$$

For a globally Lipschitz vector field, uniqueness of solutions gives an invertible finite-time flow wherever solutions exist. Liouville's general formula,

$$\det D\Phi_t(x_0) = \exp \left(\int_0^t \nabla \cdot f(x_s) \, ds \right),$$

then relates volume contraction to the divergence without identifying trajectories. A smooth taper at the anchor cutoff would place the complete realization inside this theorem.

8.5 Recovery bounds

The computed round-trip curves motivate a trajectory-level error theorem. Let

$$\kappa_T(x_t) := \|D\Psi^{-T}(x_t)\|.$$

A useful bound would expose the velocity multiplier, phase singular values, and force Lipschitz constants:

$$\|\widehat{x}_{t-T} - x_{t-T}\| \leq C_T \epsilon, \quad C_T \lesssim \prod_{s=0}^{T-1} \frac{c_s}{|\gamma| \sigma_{\min}(D_\phi G_s)}. \quad (30)$$

Given a tolerance τ , the largest T for which $C_T \epsilon \leq \tau$ would provide a certified numerical access horizon.

8.6 Observability from histories

The state-sufficiency result invites a delay-coordinate analysis. For a readout $y_t = h(x_t)$, determine when

$$(y_t, y_{t-\tau}, \dots, y_{t-(m-1)\tau})$$

embeds the reachable set, how large m must be, and how inverse conditioning scales with noise. This would connect exact-state theory to realistic partial instrumentation.

8.7 Description-level dynamics

The closure criterion [proposition 4.2](#) should be evaluated directly for the phrase readout and for task-specific alternatives. When closure fails, one can quantify the memory order required for a predictive description. When it holds in both directions, the induced macro-law inherits invertibility even though the readout is many-to-one.

8.8 Research sequence

phase domain $\longrightarrow$ invertible boundary $\longrightarrow$ recovery bound $\longrightarrow$ history observability $\longrightarrow$ domain tests.

This sequence advances the principle from exact one-step structure to certified long-horizon and domain-specific use.

9 Conclusion: the provenance of distinction

The Principle of Full Invertibility begins with a simple separation:

$$\Psi : \text{complete state} \longrightarrow \text{complete state}, \quad \mathcal{L} : \text{complete state} \longrightarrow \text{description}.$$

From that separation follows a complete audit of distinguishability.

The drift–kick realization supplies exact mechanics. Once a phase branch is fixed, velocity and position recover algebraically. The full Jacobian factors into a velocity multiplier and a phase determinant,

$$\det D\Psi = \gamma^{2N} \det D_\phi G_{r^+},$$

independently of force curvature. The phase certificate $q < 1$ identifies a unique inverse chamber for each tested update. The two-clock reduction exposes the complementary folded regime, and the three-preimage construction shows that local nonsingularity and global branch identity are distinct. The modular theorem closes the same question exactly on a finite state space.

The description layer supplies a second geometry. Its fibres collect complete states that share one retained name. The forensic fibre theorem transports those fibres backward without changing their cardinality. The closure criterion determines when a readout supports autonomous macro-dynamics. Fixed-partition entropy and float64 round trips then separate observational uncertainty from finite numerical access.

Together, these results produce a typed provenance calculus whose operations are audited separately:

$\Psi_0 : x \mapsto u$	core advance,
$\mathcal{B} : u \mapsto x^+$	boundary or controller,
$\mathcal{L} : x^+ \mapsto y$	description branch,
$\mathcal{N} : x^+ \mapsto \hat{x}^+$	numerical-state branch.

Each operation has its own injectivity or access question. Inverse branch multiplicity is the signature of noninjectivity in the preceding transition. This is the paper’s principal contribution.

The most transferable conclusion is therefore:

A state may have one past while every available description of it has many; the difference is determined by the maps through which the distinction passed.

Information loss is no longer a free-standing diagnosis. It is a located event. One declares the state, advances it, observes it, resolves its inverse branches, computes numerical access, and identifies the first operation at which the distinction is merged or becomes inaccessible. That is the provenance of distinction, and it is the operational content of full invertibility.

A Supplementary proofs

A.1 Mechanical block determinant

Let $d = 2N$, $F_r = D_{r^+}F$, and first take unit masses for notational clarity. From

$$r^+ = r + \Delta t v, \quad v^+ = \gamma v + \Delta t F(r^+),$$

the mechanical Jacobian is

$$D\Psi_{rv} := \begin{pmatrix} I_d & \Delta t I_d \\ \Delta t F_r & \gamma I_d + \Delta t^2 F_r \end{pmatrix}.$$

The Schur complement of I_d is γI_d , hence

$$\det D\Psi_{rv} = \gamma^d = \gamma^{2N}.$$

For diagonal mass matrix M , the two terms containing $\Delta t^2 M^{-1} F_r$ cancel identically.

A.2 Phase contraction proof

Write

$$K_i(\boldsymbol{\phi}) := \sum_{j \neq i} w_{ij} \sin(\phi_j - \phi_i), \quad w_{ij} := (\tilde{d}_{ij}^+)^{-1}.$$

Using $|\sin u - \sin v| \leq |u - v|$,

$$\begin{aligned} |K_i(\boldsymbol{\phi}) - K_i(\boldsymbol{\psi})| &\leq \sum_{j \neq i} w_{ij} |(\phi_j - \phi_i) - (\psi_j - \psi_i)| \\ &\leq 2 \left(\sum_{j \neq i} w_{ij} \right) \|\boldsymbol{\phi} - \boldsymbol{\psi}\|_\infty. \end{aligned}$$

Therefore the inverse fixed-point map has Lipschitz constant at most

$$q(\mathbf{r}^+) := 2 |\beta| \Delta t \max_i \sum_{j \neq i} w_{ij}.$$

When $q < 1$, Banach's theorem gives a unique fixed point on the chosen lift and geometric convergence. The forward lift $G(\boldsymbol{\phi}) = \boldsymbol{\phi} + h(\boldsymbol{\phi})$ obeys

$$\|G(\boldsymbol{\phi}) - G(\boldsymbol{\psi})\|_\infty \geq (1 - q) \|\boldsymbol{\phi} - \boldsymbol{\psi}\|_\infty,$$

which gives injectivity and the lower bi-Lipschitz bound.

A.3 Two-clock regime

For two clocks,

$$\begin{aligned} \phi_1^+ &:= \phi_1 + \omega_1 \Delta t + \frac{\beta \Delta t}{D} \sin(\phi_2 - \phi_1), \\ \phi_2^+ &:= \phi_2 + \omega_2 \Delta t + \frac{\beta \Delta t}{D} \sin(\phi_1 - \phi_2). \end{aligned}$$

Subtracting gives [equation \(14\)](#). For equal frequencies,

$$f_a(\delta) = \delta - a \sin \delta, \quad f'_a(\delta) = 1 - a \cos \delta.$$

If $|a| < 1$, then $f'_a > 0$ everywhere and the degree-one circle map is a diffeomorphism. If $|a| = 1$, the derivative is nonnegative and vanishes only at isolated points, so the map remains a strictly monotone homeomorphism but its inverse derivative is singular: at $\delta = 0 \pmod{2\pi}$ for $a = 1$, and at $\delta = \pi \pmod{2\pi}$ for $a = -1$. If $|a| > 1$, the derivative changes sign: for $a > 1$, $f'_a(0) < 0 < f'_a(\pi)$; for $a < -1$, $f'_a(\pi) < 0 < f'_a(0)$. In either case the lift has local extrema and is noninjective.

Linearization at $\delta = 0$ gives

$$\delta^+ = (1 - a)\delta + O(\delta^3).$$

Thus synchronization is hyperbolically stable for $0 < a < 2$. At $a = 2$,

$$f_2^2(\delta) := \delta - \frac{2}{3}\delta^3 + \frac{11}{30}\delta^5 + O(\delta^7),$$

so the fixed point remains locally asymptotically stable with algebraic convergence. For $a > 2$, it is unstable.

A.4 The $\gamma = 0$ collision

Choose $\mathbf{v}_2 \neq \mathbf{v}_1$ and

$$\mathbf{r}_2 := \mathbf{r}_1 - (\mathbf{v}_2 - \mathbf{v}_1)\Delta t.$$

Then

$$\mathbf{r}_1 + \mathbf{v}_1 \Delta t = \mathbf{r}_2 + \mathbf{v}_2 \Delta t = \mathbf{r}^+.$$

At $\gamma = 0$, both final velocities equal $\Delta t M^{-1} \mathbf{F}(\mathbf{r}^+, \boldsymbol{\theta}, \boldsymbol{\phi})$; the phase updates also agree. Distinct pre-states have one image.

A.5 Wall-clamp collision

Take two in-domain positions $r_1 \neq r_2 < w$ and equal velocity $v > 0$ such that both drifts overshoot w . A clamp sends both positions to w and applies the same bounce multiplier to both velocities. Every output coordinate then agrees. This proves [proposition 6.1](#).

A.6 Finite entropy identities

For a permutation $f : S \rightarrow S$,

$$p_{f(X)}(y) = p_X(f^{-1}(y)),$$

so the output probability list is a rearrangement of the input list and $H(f(X)) = H(X)$.

For a many-to-one map,

$$p_Y(y) = \sum_{x \in f^{-1}(y)} p_X(x).$$

In the uniform $g = 0$ modular system, M^2 states map evenly to M images, giving

$$H(X) = 2 \log_2 M, \quad H(Y) = \log_2 M.$$

A.7 Differential entropy transformation

Under the regularity and integrability assumptions stated before [equation \(23\)](#),

$$\rho_Y(y) := \frac{\rho_X(x)}{|\det D\Psi(x)|}, \quad x = \Psi^{-1}(y).$$

Changing variables in the differential entropy integral yields

$$h(Y) := h(X) + \mathbb{E}_X \log |\det D\Psi(X)|,$$

which is [equation \(23\)](#).

A.8 Forensic fibre cardinality

Under the assumptions of [theorem 4.1](#), Ψ^{-k} is a bijection on $\mathcal{A}$. Its restriction to $\mathcal{L}^{-1}(y) \cap \mathcal{A}$ preserves cardinality, proving

$$|\mathcal{H}_k(y)| := |\mathcal{L}^{-1}(y) \cap \mathcal{A}|.$$

B Model specification and reproducibility

B.1 Force implementation

Let

$$\Delta_{ia} = \mathbf{p}_a - \mathbf{r}_i, \quad \tilde{d}_{ia} = \|\Delta_{ia}\| + \varepsilon,$$

and

$$c_{ia} := [0.55 + 0.45 \cos(\theta_i - \theta_a)][0.60 + 0.40 \cos(\phi_i - \phi_a)].$$

The anchor force is

$$\mathbf{F}_i^{\text{anchor}} := \sum_{a=1}^M \mathbf{1}_{\{r_{\min} < \tilde{d}_{ia} < r_{\max}\}} \alpha Q_a m_i c_{ia} \frac{\Delta_{ia}}{\tilde{d}_{ia}}.$$

For $\mathbf{d}_{ij} = \mathbf{r}_i - \mathbf{r}_j$ and $\tilde{d}_{ij} = \|\mathbf{d}_{ij}\| + \varepsilon$, the implemented wave-like radial term is

$$\mathbf{F}_i^{\text{wave}} := - \sum_{j \neq i} A \left[-\frac{\sin(k\tilde{d}_{ij} + \phi_j)}{\tilde{d}_{ij}^2} + \frac{k \cos(k\tilde{d}_{ij} + \phi_j)}{\tilde{d}_{ij}} \right] \frac{\mathbf{d}_{ij}}{\tilde{d}_{ij}}.$$

The pair force is

$$\mathbf{F}_i^{\text{pair}} := \sum_{j \neq i} (-\mathbf{d}_{ij}) \left[\frac{a_{\text{att}}}{\bar{d}_{ij} + s} - \frac{a_{\text{rep}}}{\bar{d}_{ij}^2} \right].$$

For fixed external centers $\mathbf{c}_\ell$, the optional control force is

$$\mathbf{F}_i^{\text{control}} := \eta_{\text{control}} \sum_{\ell} \frac{\mathbf{c}_\ell - \mathbf{r}_i}{\|\mathbf{c}_\ell - \mathbf{r}_i\| + \varepsilon}.$$

All terms are velocity-independent. The anchor indicator is discontinuous on its shell boundaries; Jacobian claims are restricted to smooth strata.

B.2 Repository map

Path	Role
engine/psi.py	State classes, force inventory, advances, inverse-branch solvers, and diagnostics.
engine/observe.py	Anchor-basin clustering, phrase construction, geometric score, and anchor-promotion utility.
engine/discrete.py	Finite modular map, exact inverse, exhaustive permutation census, and determinant probe.
experiments/run_battery.py	Mechanical, information, phrase-fibre, boundary, and ensemble experiments.
experiments/run_deeper.py	Long-horizon, phase, wall, quantization, and recursive protocols.
experiments/run_omega.py	Two-clock, finite-state, and finite-time separation diagnostics.
evidence/thesis_verification.py	Determinant factorization, contraction certification, phase multiroot, controlled reset, fixed-partition entropy, round trips, and wall collision.
evidence/export_plot_data.py	Deterministic conversion from JSON ledgers to manuscript CSV tables.

B.3 Reproduction commands

```
python3 experiments/run_battery.py
python3 experiments/run_deeper.py
python3 experiments/run_omega.py
python3 evidence/thesis_verification.py
python3 evidence/export_plot_data.py
tectonic -X compile The_Principle_of_Full_Invertibility.tex \
  --outdir build
```

The final run used Python 3.9.6, NumPy 2.0.2, and Tectonic 0.17.0 on arm64 macOS.

B.4 State distance

Define

$$\text{wrap}(u) := \text{atan2}(\sin u, \cos u).$$

For states A, B ,

$$d_{L^2}(A, B) := \left[\|\mathbf{r}_A - \mathbf{r}_B\|_2^2 + \|\mathbf{v}_A - \mathbf{v}_B\|_2^2 + \sum_i \text{wrap}(\theta_{A,i} - \theta_{B,i})^2 + \sum_i \text{wrap}(\phi_{A,i} - \phi_{B,i})^2 \right]^{1/2}. \quad (31)$$

B.5 Phase inverse solver

The fixed-point inverse begins at

$$\boldsymbol{\phi}^{(0)} := [\boldsymbol{\phi}^+ - \omega \Delta t]_{2\pi}$$

and iterates

$$\boldsymbol{\phi}^{(k+1)} := [\boldsymbol{\phi}^+ - \omega \Delta t - \beta \Delta t \mathbf{K}(\boldsymbol{\phi}^{(k)}; \mathbf{r}^+)]_{2\pi}$$

for at most 200 iterations. Convergence requires wrapped forward residual

$$\max_i \left| \text{wrap}[G_{r^+}(\phi^{(k)}) - \phi^+]_i \right| < 10^{-15}.$$

The residual verifies one numerical root. Branch uniqueness is established separately by $q < 1$ or direct branch analysis.

B.6 Determinant probe

For $N = 2, 3, 4$ and five seeds each, the suite:

1. fixes phases away from the branch cut;
2. advances at $\gamma = 0.982$, $\beta = 0.08$, and $\Delta t = 0.21$;
3. forms a central finite-difference Jacobian with step 2×10^{-7} ;
4. computes $D_\phi G$ analytically;
5. compares the full determinant with $\gamma^{2N} \det D_\phi G$.

The maximum relative discrepancy is 2.01×10^{-8} .

B.7 Multiroot construction

The default $N = 8$, seed-22 field is advanced through step 9. Three Newton initializations converge to three roots of the wrapped phase equation. For each root, [equations \(10b\)](#) and [\(10c\)](#) reconstructs the complete pre-state, which is advanced once to verify the common target. Alternative pre-states lie 7.386 and 6.738 from the generating pre-state and return to the target within 1.50×10^{-13} and 1.33×10^{-13} .

B.8 Round-trip ensemble

The endpoint-inversion ensemble uses:

- $N = 8$ tokens and seeds 21–32;
- $\gamma \in \{1, 0.982\}$;
- $T \in \{40, 80, 100, 120, 160, 200\}$;
- default $\beta = 0.08$, $\Delta t = 0.35$;
- no walls and no directional drift;
- float64 arithmetic.

Only trajectories with $q < 1$ at every step and maximum wrapped forward phase residual below 10^{-12} enter the displayed summaries. All $\gamma = 1$ trajectories are certified through 200 steps. For $\gamma = 0.982$, 11 are certified through 160 steps and 8 through 200.

B.9 Phase-transport and reset protocols

Two tokens are coupled for four steps with translational forces disabled. Coupling is then set exactly to zero for 300 steps. Equal-frequency and unequal-frequency traces are compared with [equation \(26\)](#).

For the reset experiment, one phase state is cloned at separations 0.3, 1, 3, 24. A no-reset control and a $\phi_1 \leftarrow 0$ arm are advanced once. Their partner-phase difference is compared with [equation \(27\)](#).

B.10 Phrase-fibre census

For each of 160 seeds, 14 tokens advance 90 steps without walls. Clusters are assigned to the nearest anchor within radius 1.15; remaining tokens are grouped as FREE. The census stores the phrase and complete-state fingerprint. It measures fibre multiplicity independently of the auxiliary geometric score.

B.11 Fixed-partition entropy

The fixed-partition diagnostic declares one set of position and velocity edges with overflow bins and 12 equal circular phase bins before evolution. It computes empirical marginal Shannon entropy per coordinate across 120 ensemble members and reports their sum. The quantity is explicitly a sum of marginals, not a joint entropy.

B.12 Numerical scope

1. Hard force cutoffs define smooth strata separated by discontinuity surfaces.
2. A converged root residual verifies one branch, not global uniqueness.
3. The 200-iteration cap and convergence flag are retained in the ledger.
4. The finite-time two-orbit separation diagnostic is not a Lyapunov spectrum.
5. Rounded random-image uniqueness is not used as evidence of injectivity.
6. Kinetic-energy drift is not used as evidence about a total energy integral.
7. The supplemental suite imports the same engine and is a self-consistency audit.

B.13 Public evidence package

The public repository contains the manuscript source, executable audit, machine-readable ledgers, exact plot tables, checksums, and reproduction instructions.

C Result ledger

C.1 Proved results

1. If $\gamma \neq 0$ and the phase map at the advanced position is bijective, the wall-free one-step map has the inverse in [equation \(10\)](#).
2. On smooth strata,

$$\det D\Psi = \gamma^{2N} \det D_\phi G_{r^+}.$$

3. The mechanical block is bijective for every velocity-independent force when $\gamma \neq 0$.
4. The phase inverse is contraction-certified when $q(r^+) < 1$.
5. The two-clock difference obeys [equation \(14\)](#); its map is a circle diffeomorphism for $|a| < 1$ and folds for $|a| > 1$.
6. The finite modular map is bijective exactly when $\gcd(g, M) = 1$.
7. A finite bijection preserves discrete Shannon entropy.
8. Differential entropy transforms by the expected log-Jacobian on a smooth bijective stratum.
9. The history-fibre cardinality equals the reachable observation-fibre cardinality under a bijective advance.
10. A deterministic description-level map exists exactly under the fibre-compatibility condition [equation \(22\)](#).
11. The wall clamp in [proposition 6.1](#) is many-to-one.

C.2 Computed results

Result	Status	Value
Full determinant probe	COMPUTED	15 live-map probes; maximum relative discrepancy 2.01×10^{-8} .
Default phase multiroot	COMPUTED	Three complete pre-states; alternative distances 7.386, 6.738; forward residuals below 1.6×10^{-13} .
Finite permutation census	COMPUTED	63,001/63,001 images for $(M, g) = (251, 3)$; 128/256 for $(16, 2)$.
Phrase fibres	COMPUTED	160 states, 103 phrases, 31 repeated phrases, 68/68 checked shared-phrase pairs distinct.
Complete versus partial state	COMPUTED	Complete 30-step recovery 1.31×10^{-10} ; zero-velocity 15.35; zero-phase 25.37.
Float64 round trip at $T = 200$	COMPUTED	Median 3.02×10^{-6} for $\gamma = 1$ (12 seeds); 3.04×10^{-3} for $\gamma = 0.982$ (8 certified seeds).
Fixed-partition marginal entropy	COMPUTED	$\gamma = 1$: 84.62 $\rightarrow$ 110.40 bits; $\gamma = 0.982$: 84.62 $\rightarrow$ 99.26 bits at $t = 150$.
Controlled reset response	COMPUTED	0.32266 rad at $D = 0.3$, 0.004047 rad at $D = 24$, matching $\tilde{D}^{-1}$.
Wall collision	COMPUTED	Pre-state distance 0.2, final distance 0.

C.3 Open mathematical objects

1. A nontrivial invariant domain on which the full phase-coupled map is globally bijective.
2. A sharp spectral characterization of the phase-map bijectivity chamber.
3. An invertible bounded realization with periodic or specular boundaries.
4. A trajectory-level round-trip error bound with explicit force and phase condition factors.
5. Delay-coordinate observability bounds for partial token measurements.
6. A complete task-specific description code and its induced closure properties.
7. A physical domain in which the declared complete state and parameters are independently calibrated.

C.4 Notation

Symbol	Meaning
$\mathcal{X}$	Complete modeled state space.
$\mathcal{A}$	Admissible domain on which a stated theorem or experiment is evaluated.
x	One complete modeled state.
Ψ	State-advance map.
$\mathcal{L}$	Description or observation map.
$\mathcal{Y}$	Description space.
$\mathcal{L}^{-1}(y)$	Observation fibre of description y .
$\mathbf{r}, \mathbf{v}$	Token positions and velocities.
$\boldsymbol{\theta}, \boldsymbol{\phi}$	Directional and phase coordinates.
γ	Velocity multiplier.
β	Phase-coupling strength.
G_r	Phase update at fixed advanced positions.
$q(\mathbf{r})$	Sufficient phase-inverse contraction certificate.
$\mathcal{H}_k(y)$	Histories compatible with description y .

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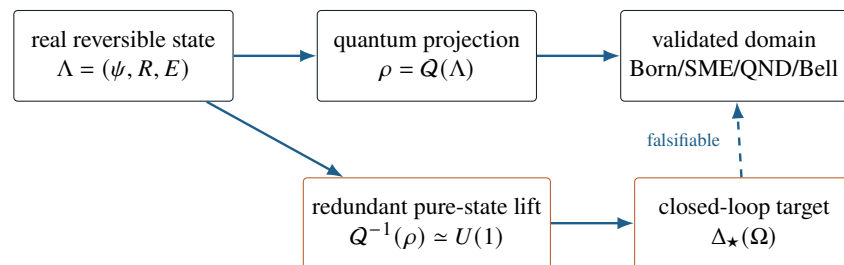
Emergent Predictive Representation

Complex structure, quantum sufficiency, and a one-parameter Hopf-fibre test

Logan Voss

Voss Dynamics | Working mathematical thesis and experimental protocol

21 August 2026



Evidence statement. The conditional characterization theorems and reported algebraic identities are proved within their stated assumptions. The numerical results are internal model checks or explicitly labelled synthetic software validations. No laboratory deviation from quantum mechanics is reported. VD-Hopf-1 is a phenomenological ansatz with a memoryless reduced-qubit null at $\varepsilon = 0$, not an established law of nature.

Abstract

This thesis separates an explanatory reconstruction of quantum structure from a test of reduced-state predictive sufficiency. Conditionally on irreducible Euclidean Jordan geometry, a normalized nontrivial stationary real detector reference has a two-dimensional cyclic span. A binary face of Peirce degree d has an unnormalized reversible QND-filter algebra of dimension $1 + d(d - 1)/2$. An explicit accessibility and isomorphism axiom between these spaces forces $d = 2$, selecting the complex Hermitian family. The result is dimension-forcing and conditional; the decisive response-completeness premise is not derived.

For a pure qubit, the redundant lift $\psi \mapsto \rho = \psi\psi^\dagger$ has a $U(1)$ Hopf fibre. The ideal QND chart $s = \operatorname{artanh} z$ yields $dx_\varphi = a dY$ and $dc_\varphi = 0$, showing that the transverse QND invariant is already encoded in ρ . VD-Hopf-1 therefore postulates instead that a gauge-invariant closed-path holonomy Γ_{rel} survives as physical history memory and enters a later record through

$$dY_t = a[z_t + \varepsilon(1 - z_t^2) \sin \Gamma_{\text{rel},t}]dt + dW_t.$$

Mirror loops give a leading orientation-odd sine law. The memoryless reduced-qubit null is $\varepsilon = 0$.

All quantum-limit checks and synthetic statistics are internal regressions. No memory carrier, mixed/composite instrument, or beyond-quantum result is derived or observed. The experimental deliverable is a draft bounded-model protocol: a positive result would first show unexplained history dependence relative to specified reduced-qubit, qutrit–cavity, and finite-memory models.

Primary result The defensible mathematical refinement is a *fibre distinction*: the QND transverse invariant is hidden from one record but encoded in the quantum state; a normalized pure-spinor lift has a redundant $U(1)$ fibre. VD-Hopf-1 separately postulates a gauge-invariant holonomy memory and asks whether a protocol-fixed matched-state variation is nonzero.

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1 Question, standard of proof, and the six courts

1.1 The thesis in one equation

Let Λ denote a complete candidate state, A an intervention, and O its outcome. Because “EPR” conventionally denotes Einstein–Podolsky–Rosen in quantum foundations, this manuscript writes *Emergent Predictive Representation* in full and calls the concrete ansatz VD-Hopf-1. The proposition is that a map

$$\rho = Q(\Lambda) \quad (1)$$

reproduces the successful quantum domain but may be predictively lossy. Conditional mutual information requires a preparation prior. For an admissible prior μ on Λ and bounded intervention class $\mathcal{A}$, define

$$\mathcal{E}(\mu, \mathcal{A}) := \sup_{a \in \mathcal{A}} I_\mu(\Lambda; O \mid \rho, A = a), \quad (2)$$

and, if needed, maximize over a declared admissible prior family $\mathfrak{M}$. A model-level separating pair obeys

$$\Lambda_+ \neq \Lambda_-, \quad Q(\Lambda_+) = Q(\Lambda_-) = \rho, \quad P(O \mid a, \Lambda_+) \neq P(O \mid a, \Lambda_-). \quad (3)$$

Because Λ is not directly observed, [equation \(2\)](#) is prior-dependent and theoretical. The total-variation objective later is the primary prior-free model criterion, and the estimable witness replaces Λ by randomized preparation histories. A same-reduced- ρ pair establishes reduced-state insufficiency; it does not by itself exclude an enlarged standard quantum state.

1.2 Claim discipline

Four statuses are used throughout:

PROVED

A mathematical consequence of explicitly listed assumptions.

COMPUTED

A symbolic or numerical identity reproduced by code and tests.

CANDIDATE

A specified law that is consistent enough to test but is not inferred from physical data.

OPEN

A physical or novelty claim requiring experiment, stronger axioms, or independent literature confirmation.

No synthetic dataset can establish that Nature implements a candidate mechanism. No fit to an unsealed target counts as a prediction. A reconstruction of standard quantum mechanics is not evidence against standard quantum mechanics.

1.3 Court ledger

Table 1: The six courts and their present status. The ledger deliberately separates conditional mathematics, baseline checks, candidate definitions, and open physical work.

Court	Required object	Result in this thesis
1	Small admissible real geometry under strengthened reversibility, stationary reference, QND calibration, covariance, causality, mixing, and no-signalling constraints.	CONDITIONAL: reference and filter-stabilizer lemmas; dimension-forcing accessibility/completeness axiom remains.
2	A map $Q : (\xi, R, E) \mapsto \rho$, with phase structure identified rather than renamed.	PARTIAL: exact redundant pure-spinor lift and QND chart; no derived physical fibre ontology or canonical mixed-state fibre.
3	Born probabilities, unitaries, trajectories, phase backaction, dephasing, Rabi, Zeno, two-qubit correlations, CHSH, and no-signalling.	BASELINE CHECK: exact at $\varepsilon = 0$; not an independent derivation of the full instrument calculus.
4	Lowest-order departures surviving declared symmetries.	CANDIDATE: one chosen basis element; no complete mixed/composite instrument.
5	Two deeper states with the same matched projected state and different predictions.	MODEL-CONDITIONAL: same reduced ρ , opposite holonomy label, and nonzero ansatz drift; physical memory carrier is postulated.
6	Frozen target, likelihood, calibration boundary, controls, and failure rule.	DRAFT: synthetic Gaussian summary validation; bounded process tensor, qutrit–cavity null, path score, thresholds, and external commitment remain open.

2 Court 1: admissible state geometry

2.1 Operational state before ontology

Let h be a complete available preparation history. Two histories are operationally equivalent when no allowed future intervention distinguishes them:

$$h \sim h' \iff P(o \mid \text{do}(a), h) = P(o \mid \text{do}(a), h') \quad \forall a, o. \quad (4)$$

An operational state is the class $[h]$. Independent classical randomization makes the normalized state set convex, and effects are affine probability functionals. This is standard operational-probabilistic structure, not yet quantum theory.

The finite-dimensional homogeneous self-dual (HSD) cone assumption is substantial. By the Koecher–Vinberg correspondence, an HSD cone is the cone of squares of a Euclidean Jordan algebra (EJA) [1–3]. Modern work seeks operational replacements for self-duality [4]; this thesis does not claim to have removed that premise.

Axiom 2.1 (Jordan domain). The finite-dimensional unnormalized operational states form an irreducible homogeneous self-dual cone of rank $r \geq 2$. Equivalently, they are represented by an irreducible EJA J with binary faces.

The Jordan–von Neumann–Wigner classification contains real, complex, and quaternionic Hermitian matrix algebras, spin factors, and the exceptional 3×3 octonionic algebra [3, 5]. The task is therefore not to rediscover that list, but to identify an observation principle that selects one member.

2.2 Measure preservation excludes positive-measure attraction

Proposition 2.2 (No positive-measure attraction in a measure-preserving flow). *Let X be a metric space with a finite regular Borel measure μ , and let Ψ_t be an invertible measure-preserving flow. If $A \subset X$ is closed with $\mu(A) = 0$, there is no measurable B with $\mu(B) > 0$ that is uniformly attracted to A , meaning that for every open neighborhood $N \supset A$ there is t_N such that $\Psi_t(B) \subset N$ for all $t \geq t_N$.*

Proof. Regularity and $\mu(A) = 0$ give an open $N \supset A$ with $\mu(N) < \mu(B)$. Uniform attraction would eventually imply $\Psi_t(B) \subset N$, whereas measure preservation gives $\mu(\Psi_t B) = \mu(B) > \mu(N)$, a contradiction. $\square$

Invertibility by itself does not imply this result: $\dot{x} = -x$ has an invertible finite-time flow and contracts. The Voss architecture therefore needs measure preservation, Hamiltonian/unitary structure, or another explicit conservation law in addition to reversibility if damping and stochastic collapse are to be reduced descriptions only. Under that strengthened baseline, a complete candidate state retains system, reference, and environment variables,

$$\Lambda = (\xi, R, E), \quad \Lambda_{t+\Delta t} = \Psi_{\Delta t}(\Lambda_t), \quad \Psi_{\Delta t}^{-1} \text{ exists.} \quad (5)$$

Unitary system–field dilation is an established way to realize this logic for quantum filters [6, 7]. Reversibility alone does not determine Ψ .

2.3 Smallest symmetry-forced generator class

Court 1 asks for the smallest F , not an invented damping field. On a finite-dimensional real complete state X with stationary quadratic calibration $\|X\|^2$, assume a local linear control law

$$\dot{X} = A(u)X.$$

Stationarity for every X gives

$$\frac{d}{dt}\|X\|^2 = X^\top [A(u)^\top + A(u)]X = 0,$$

and therefore

$$A(u)^\top = -A(u). \quad (6)$$

Thus the minimal closed linear class is orthogonal transport,

$$F_0(X, u) = A(u)X, \quad A(u) \in \mathfrak{so}(N),$$

with no contraction coefficient. The elementary spinor requires $X_\psi \in S^3 \subset \mathbb{R}^4$; the reference theorem below supplies a real rotation block; measurement requires additional environment/field coordinates. Conditioning or tracing those coordinates can yield a stochastic master equation, but reduced damping is not inserted into F_0 .

This is a classification only within the declared linear stationary class. General reversible nonlinear dynamics may be Hamiltonian, $\dot{X} = \mathbb{J}\nabla H(X, u)$, and are not uniquely fixed by the present axioms. The thesis therefore uses the smallest linear lift as the baseline and parameterizes every proposed departure explicitly.

2.4 Minimal real reference

Let W be a finite-dimensional real detector-reference space with its calibrated covariance inner product. Let $P \in W$ be a centered real pointer and let changing the reference origin act by a strongly continuous one-parameter group R_s .

Axiom 2.3 (Stationary cyclic reference). P is normalized, $\|P\| = 1$; R_s preserves the covariance inner product; the action is nontrivial on P , so $R_s P \neq P$ for some s ; and the cyclic subspace

$$W_P = \text{span}\{R_s P : s \in \mathbb{R}\}$$

is irreducible under R_s .

Theorem 2.4 (Minimal real reference). Under axiom 2.3, $\dim W_P = 2$. There are orthonormal P, Q and $\omega > 0$ such that

$$R_s P = P \cos(\omega s) + Q \sin(\omega s). \quad (7)$$

The normalized orbit is S^1 , and $J_R := A/\omega$ obeys $J_R^2 = -I$ on W_P .

Proof. Stationarity gives $R_s \in O(W)$. Strong continuity gives $R_s = e^{sA}$, and differentiating $R_s^\top R_s = I$ yields $A^\top = -A$. Every real skew-symmetric operator is an orthogonal direct sum of zero blocks and two-dimensional rotation blocks. Nontriviality excludes a zero block; irreducibility excludes more than one block. Hence W_P is one rotation plane. Choosing $Q = AP/\omega$ gives equation (7) and $J_R^2 = -I$. $\square$

The theorem is elementary representation theory. Its relevance is physical only if W_P is independently calibrated and its response to the system is not postulated to have the desired dimension.

2.5 Binary Jordan faces and their unnormalized QND-filter stabilizer

Let p, q be orthogonal primitive idempotents. Their rank-two face has Peirce decomposition

$$J[p, q] = \mathbb{R}p \oplus \mathbb{R}q \oplus J_{pq}, \quad d := \dim J_{pq}. \quad (8)$$

Every rank-two EJA is a spin factor. Its unnormalized cone can be written as the Lorentz cone

$$\mathcal{L}_{d+2} = \{(t, z, \mathbf{c}) : t \geq 0, t^2 \geq z^2 + \|\mathbf{c}\|^2, \mathbf{c} \in \mathbb{R}^d\}, \quad (9)$$

with QND outcome rays $\ell_{\pm} = \mathbb{R}_{\geq 0}(1, \pm 1, \mathbf{0})$.

Lemma 2.5 (Unnormalized QND-filter stabilizer). *Modulo global cone dilation, the connected Lie algebra of linear Lorentz-cone automorphisms preserving each ray ℓ_+ and ℓ_- is*

$$\mathfrak{f}_{pq} \cong \mathbb{R}K_{pq} \oplus \mathfrak{so}(d), \quad \dim \mathfrak{f}_{pq} = 1 + \frac{d(d-1)}{2}. \quad (10)$$

Proof. The connected cone automorphism algebra is $\mathbb{R}I \oplus \mathfrak{so}(1, d+1)$. Requiring an infinitesimal generator to map both null rays into themselves removes boosts mixing (t, z) with $\mathbf{c}$ and removes null rotations. After quotienting the central dilation, the remaining blocks are the single t - z boost K_{pq} and arbitrary rotations of $\mathbf{c}$, namely $\mathfrak{so}(d)$. The dimension follows. $\square$

The word “reversible” here means invertible as an unnormalized cone order-automorphism. The boost K_{pq} changes the order unit and is a probabilistically reversible filter/measurement branch, not a deterministic reversible channel on normalized states. Quotienting $\mathbb{R}I$ is appropriate only for normalized conditional-state transformations; the overall branch scale remains relevant to raw outcome probability and must be calibrated separately. The normalized deterministic stabilizer would retain only $\mathfrak{so}(d)$, and the dimension argument below would not follow.

2.6 Conditional complex-scalar characterization

Detector reference directions need not automatically equal system generator directions. The bridge is isolated as an axiom rather than hidden in the prose.

Axiom 2.6 (Accessible, faithful, and generator-complete filter response). Every vector in the two-dimensional linear span W_P , not merely the tangent to its one-parameter orbit, is assumed to be an independently accessible infinitesimal detector perturbation. “Complete” means direct first-order linear spanning, not reachability generated by sequential controls or Lie brackets. For every binary face these perturbations induce a linear response

$$\Phi_{pq} : W_P \longrightarrow \mathfrak{f}_{pq}$$

that is injective (no accessible detector direction is idle) and surjective (no unnormalized reversible QND-filter generator is inaccessible).

Theorem 2.7 (Conditional detector/filter complex-scalar characterization). *Let J satisfy [axiom 2.1](#). If [axioms 2.3](#) and [2.6](#) hold for every binary face, then*

$$J \simeq H_r(\mathbb{C})_{\text{sa}}.$$

For rank two, the state space is the ordinary complex qubit.

Proof. [Theorem 2.4](#) gives $\dim W_P = 2$. Faithfulness and completeness make Φ_{pq} an isomorphism, so $\dim \mathfrak{f}_{pq} = 2$. By [lemma 2.5](#),

$$2 = 1 + \frac{d(d-1)}{2}.$$

The only nonnegative integer solution is $d = 2$. The classification of irreducible EJAs then leaves the complex Hermitian series; the rank-two spin factor at $d = 2$ is $H_2(\mathbb{C})_{\text{sa}}$. $\square$

The proof uses only vector-space dimension. A generic bijection Φ_{pq} need not be isometric, need not map the detector quarter-turn J_R to the canonical system complex structure, and can map a detector circle to an ellipse. Any stronger identification of the physical meaning of i requires an additional equivariance or calibration theorem.

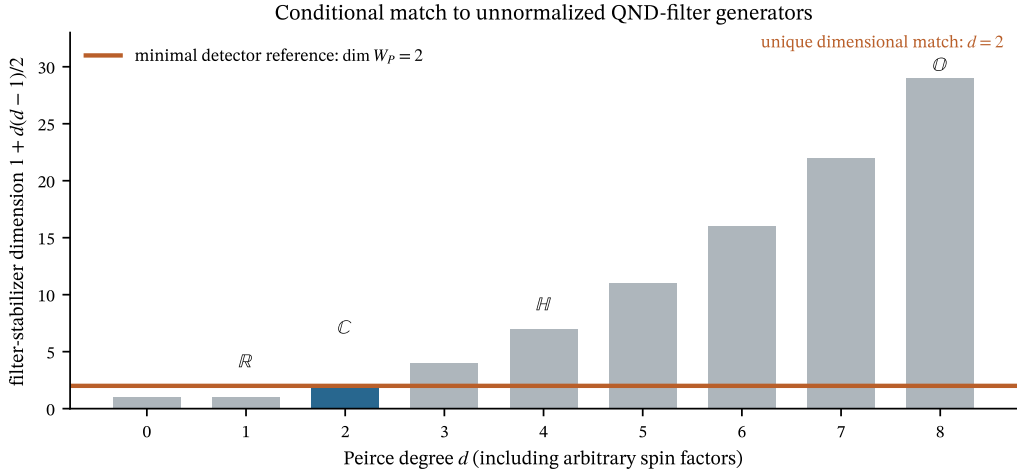


Figure 1: The detector-reference span and unnormalized QND-filter stabilizer dimension agree only at Peirce degree $d = 2$. Arbitrary integer degrees, including rank-two spin factors, are shown. The visual does not weaken the logical qualification: accessibility, injectivity, and surjectivity of Φ_{pq} are assumptions.

Novelty boundary. The EJA classification, Koecher–Vinberg theorem, Peirce geometry, and Lorentz-cone groups are established mathematics [3, 5]. Local tomography plus Jordan structure and a qubit already selects ordinary complex quantum theory under additional factorizable-self-duality and composite-system assumptions [8]; energy observability and dynamical correspondence provide neighboring complex selectors [9, 10]; reversible filters are also established reconstruction machinery [11]. The potentially distinct contribution here is only the *specific conditional dimension-matching criterion* between a minimal real detector-reference span and the complete unnormalized QND-filter algebra. The one-parameter orbit itself has a one-dimensional tangent; treating both vectors in W_P as independently accessible perturbations is part of [axiom 2.6](#), not a consequence of [theorem 2.4](#). Until peer-reviewed novelty audit and derivation of that axiom are complete, the result must not be described as “observation alone proves $\mathbb{C}$.”

2.7 Causality, mixing, and no-signalling

Operational causality requires normalized probabilities for every deterministic test and forbids dependence on future interventions. Classical randomization is affine at the level of probability measures on Λ :

$$\mu = \lambda\mu_1 + (1 - \lambda)\mu_2 \implies P(o \mid a, \mu) = \lambda P(o \mid a, \mu_1) + (1 - \lambda)P(o \mid a, \mu_2).$$

At $\varepsilon = 0$, the complex quantum representation and tensor product recover standard no-signalling. If different preparation histories with the same operational state produced different future laws, the model would be preparation contextual in the generalized sense [12]; its bipartite extension is not licensed by single-system equations. [Section 5.5](#) states the operational constraint relevant to the Gisin–Polchinski signalling mechanism [13, 14].

3 Court 2: explicit quantum representation and redundant lift

3.1 From a real rotation plane to a complex structure

On the detector reference plane, $J_R^2 = -I$. The notation

$$u + iv \longleftrightarrow uI + vJ_R$$

is therefore a compression of a real two-dimensional rotation. This explains how a complex structure can be represented by real relational geometry; it does not show that every physically allowed transformation must commute with the same J_R .

Given [theorem 2.7](#), an elementary pure state is a rank-one projector that *can be represented redundantly* by a normalized two-component complex spinor $\psi = (\alpha, \beta)^T \in S^3 \subset \mathbb{R}^4$. The EJA theorem yields the projector/ray, not a physical point of S^3 . The representative map is

$$Q(\psi) = \psi\psi^\dagger = \frac{1}{2}(I + \mathbf{r} \cdot \boldsymbol{\sigma}). \quad (11)$$

This is the Hopf fibration $S^1 \hookrightarrow S^3 \rightarrow S^2$ [15]. It is invariant under

$$\psi \mapsto e^{i\chi} \psi. \quad (12)$$

Thus the fibre $Q^{-1}(\rho)$ within the normalized pure-spinor lift is $U(1)$.

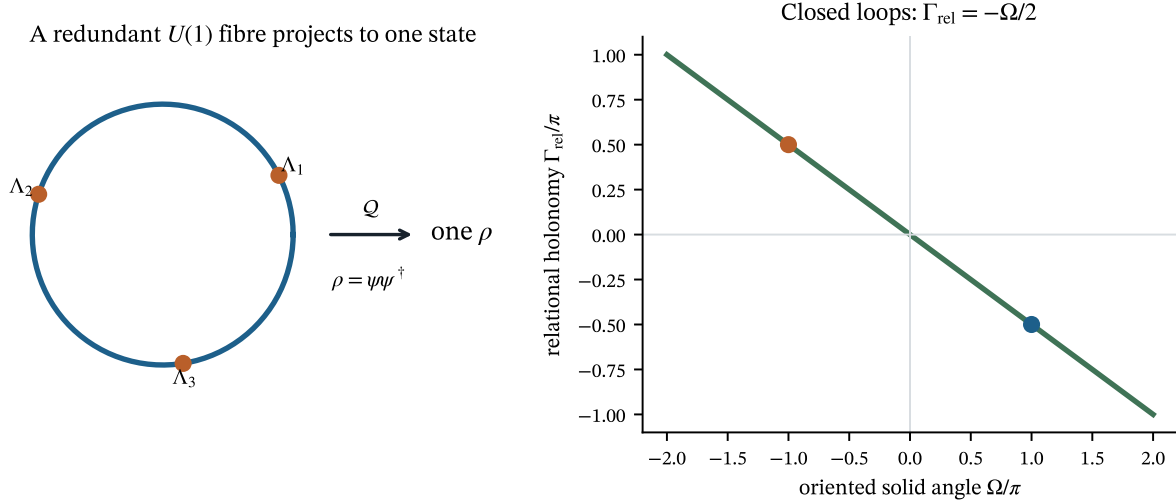


Figure 2: Left: many spinors on one $U(1)$ fibre project to the same density matrix. Right: the natural spin- $\frac{1}{2}$ connection produces closed-loop holonomy $\Gamma_{\text{rel}} = -\Omega/2$. Standard quantum mechanics predicts no system-only distinction after the same ray is restored; any physical reference must be included in the declared bounded enlarged-quantum null.

The Hopf representation is standard, and retaining the full S^3 geometry or debating physical significance of the discarded phase is not new [15, 16]. The distinct candidate contribution sought here is narrower: formulate a matched protocol-level variation objective, isolate a gauge-invariant closed-history label, and freeze a future nominally QND record ansatz that standard enlarged quantum models must try to explain.

3.2 Exact relational chart

Choose a local gauge with $\alpha > 0$ and define

$$\frac{\beta}{\alpha} = e^{-s+i\vartheta}, \quad s \in \mathbb{R}, \quad \vartheta \in S^1. \quad (13)$$

Then

$$\mathbf{r}(s, \vartheta) = (\text{sech } s \cos \vartheta, \text{sech } s \sin \vartheta, \tanh s), \quad (14)$$

and

$$\rho(s, \vartheta) = \frac{1}{2} \begin{pmatrix} 1 + \tanh s & \text{sech } s e^{-i\vartheta} \\ \text{sech } s e^{i\vartheta} & 1 - \tanh s \end{pmatrix}. \quad (15)$$

The poles are obtained as $s \rightarrow \pm\infty$. Mixed states are convex mixtures, or equivalently

$$\rho_{\nu, s, \vartheta} = \frac{1}{2} [I + \nu \mathbf{r}(s, \vartheta) \cdot \boldsymbol{\sigma}], \quad 0 \leq \nu \leq 1,$$

but this chart is nonunique at $\nu = 0$, and a mixed density operator has no canonical $U(1)$ Hopf fibre. Purification, Uhlmann, and interferometric holonomies are different structures [17, 18].

The complete real candidate coordinate can be written

$$\Lambda = (s, \vartheta, \chi, E), \quad \psi(\Lambda) = \frac{e^{i\chi}}{\sqrt{1 + e^{-2s}}} \begin{pmatrix} 1 \\ e^{-s+i\vartheta} \end{pmatrix}. \quad (16)$$

The explicit effective map used here is

$$Q(s, \vartheta, \chi, \nu) := \rho_{\nu, s, \vartheta}. \quad (17)$$

At $\nu = 1$ this is the Hopf projection of [equation \(16\)](#); for $\nu < 1$, χ is only a redundant coordinate attached by this parameterization, not a canonical mixed-state fibre.

In the notation $\Lambda = (\xi, R, E)$, one may take

$$\xi = s, \quad R = (e^{i\vartheta}, e^{i\chi}), \quad E \supset \nu,$$

where the first circle is the component-relative phase relation, the second is the mathematical Hopf fibre, and ν records reduced purity. This is a coordinate decomposition, not a claim that both circles are independently observable. The later candidate replaces bare χ by the gauge-invariant closed-history memory $\Gamma_{\text{rel}}[C]$.

The effective state $\rho = Q(\Lambda)$ forgets χ . In relational language, ϑ is the phase between spinor components; χ is the phase of the whole spinor relative to an external reference. Quantum reference-frame theory warns that a physical reference is itself a quantum system and that only invariant relational quantities are observable [19, 20]. This warning is central to the null model.

Accordingly, VD-Hopf-1 below is restricted to conditioned pure, efficient trajectories ($\nu = 1$) and is only a phenomenological record ansatz. A hardware protocol must either enforce a preregistered near-purity/no-jump slice with quantified selection bias or first supply a mixed-state or purification-level holonomy law. The present thesis does not contain that extension.

3.3 The ideal QND chart

For an efficient phase-sensitive Z-QND measurement, take the normalized Itô model

$$dY_t = a \cos \varphi z_t dt + dW_t, \tag{18}$$

$$dz_t = a \cos \varphi (1 - z_t^2) dW_t, \tag{19}$$

$$d\vartheta_t = a \sin \varphi dW_t + a^2 \sin \varphi \cos \varphi z_t dt. \tag{20}$$

Factors depend on measurement-strength convention; the identities below are convention-consistent. Circuit-QED quantum Bayesian and filtering treatments contain precisely the split between informational and phase backaction controlled by homodyne angle [21–23].

Let $s = \text{artanh } z$. Itô's lemma gives

$$ds_t = a \cos \varphi dY_t, \quad d\vartheta_t = a \sin \varphi dY_t. \tag{21}$$

Define

$$x_\varphi = s \cos \varphi + \vartheta \sin \varphi, \tag{22}$$

$$c_\varphi = -s \sin \varphi + \vartheta \cos \varphi. \tag{23}$$

Then

$$\boxed{dx_\varphi = a dY_t, \quad dc_\varphi = 0.} \tag{24}$$

Equivalently, $\chi_{\text{QND}} = \vartheta - \tan \varphi \text{artanh } z = c_\varphi / \cos \varphi$ is conserved where $\cos \varphi \neq 0$. The supplied blind symbolic program recovered both the invariant and the observable quotient with zero Itô residual; its control search also recovered the local D-optimal axis $|n_z| = 1/\sqrt{3}$. These are checks of known quantum dynamics, not beyond-quantum evidence.

The construction is local: ϑ is unwrapped on a chart, whereas the physical phase is periodic, and the poles require separate coordinates. It is not a global orthogonal coordinate system on the Bloch sphere.

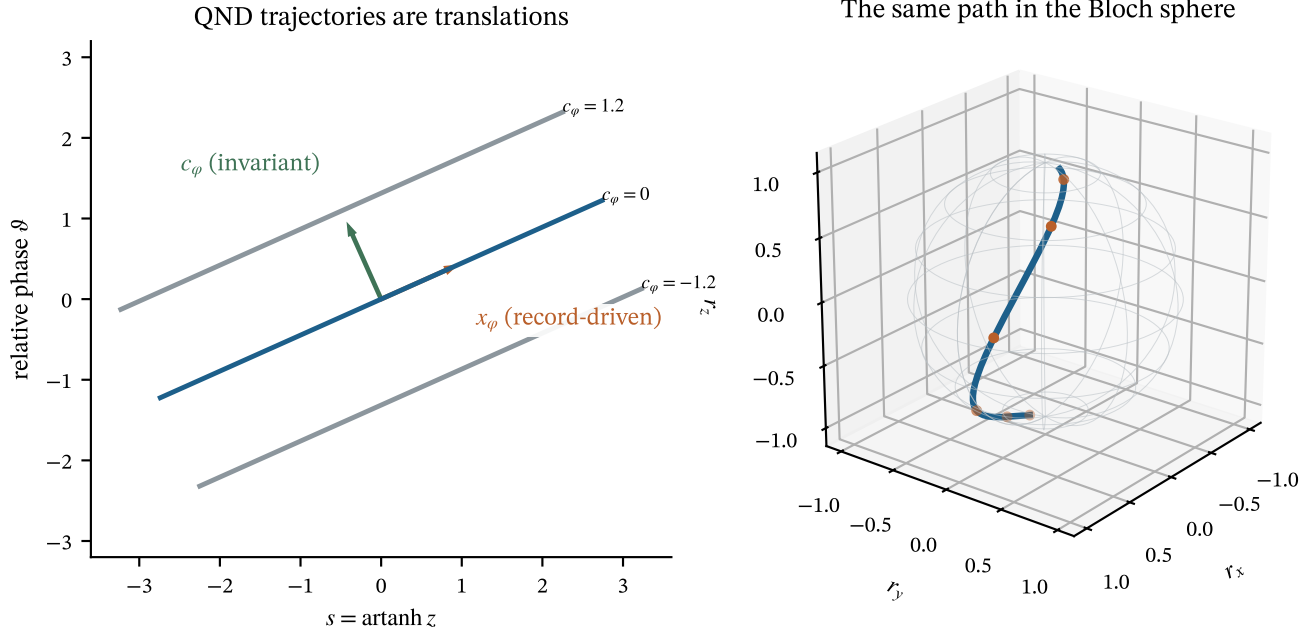


Figure 3: The nonlinear pure-state QND trajectory becomes a straight translation in (s, ϑ) . The record determines position along a line; c_φ labels which line. The right panel maps one line back to the Bloch sphere.

3.4 Fibre distinction theorem

Theorem 3.1 (QND transverse information is not fibre information). *On any chart where $(s, \vartheta) \leftrightarrow \rho$ is one-to-one, c_φ is a function of ρ and the known setting φ . Therefore two states satisfying $Q(\Lambda_1) = Q(\Lambda_2)$ cannot differ in c_φ . By contrast, they may differ in the Hopf coordinate χ .*

Proof. Equation (15) recovers $z = \operatorname{Tr}(\rho \sigma_z)$, hence $s = \operatorname{artanh} z$, and recovers $\vartheta = \arg \rho_{10}$ away from the poles. Thus $c_\varphi = -s \sin \varphi + \vartheta \cos \varphi$ is fixed by ρ and φ . Equations (11) and (12) show that χ is not. $\square$

This theorem changes the experimental target. The QND invariant is important for control and observability, but assigning it to a reference cannot by itself make ρ predictively incomplete. A Court-5 variable must distinguish points in a fibre of Q ; the smallest connected fibre in the chosen normalized pure-spinor lift is $U(1)$.

4 Court 3: recovery of validated quantum limits

4.1 Born effects and an explicit logical boundary

Once the complex positive cone is adopted, every affine effect has the form

$$e_M(\rho) = \operatorname{Tr}(\rho M), \quad 0 \leq M \leq I. \quad (25)$$

Sharp calibration to a primitive outcome gives $M = \Pi_k$, hence

$$P(k \mid \rho) = \operatorname{Tr}(\rho \Pi_k). \quad (26)$$

This is the Born rule in the reconstructed representation. Gleason-type results explain the measure structure, with generalized effects including the qubit case [24, 25]. It would be circular to call equation (26) a derivation from basin volumes if affinity or effect structure had already inserted the quantum probability functional. The present result is therefore: *conditional scalar reconstruction plus affine effects recovers Born probabilities*; it is not a non-quantum dynamical proof of Born's rule.

4.2 Unitary and continuous-measurement dynamics

For $H = \frac{\hbar}{2} \boldsymbol{\omega} \cdot \boldsymbol{\sigma}$,

$$\dot{\rho} = -\frac{i}{\hbar} [H, \rho] \iff \dot{\mathbf{r}} = \boldsymbol{\omega} \times \mathbf{r}. \quad (27)$$

This is a reversible $SO(3)$ action on the Bloch ball and an $SU(2)$ lift on S^3 . A standard efficient diffusive measurement of Z can be written

$$d\rho = -\frac{i}{\hbar} [H, \rho] dt + \Gamma_m \mathcal{D}[\sigma_z] \rho dt + \sqrt{\eta \Gamma_m} \mathcal{H}[e^{-i\varphi} \sigma_z] \rho dW, \quad (28)$$

where

$$\mathcal{D}[L] \rho = L \rho L^\dagger - \frac{1}{2} \{L^\dagger L, \rho\}, \quad \mathcal{H}[L] \rho = L \rho + \rho L^\dagger - \text{Tr}[(L + L^\dagger) \rho] \rho.$$

The innovations filter follows from quantum conditional expectation and system–field dilation [6, 7]. Experimental quantum trajectories and variable quadrature backaction are established [23, 26–28].

4.3 Dephasing, Rabi dynamics, and the Zeno limit

Unobserved measurement dephases transverse coherence,

$$\rho_{01}(t) = \rho_{01}(0) e^{-\Gamma_\phi t},$$

while a transverse drive produces Rabi rotation. For strong Z -measurement dephasing $\Gamma_m \gg \Omega_R$, transitions generated by a noncommuting drive are suppressed: the quantum Zeno limit. Circuit-QED trajectory theory derives both quantum jumps and this suppression [22]. The computations used here exponentiate the Bloch generator, rather than infer the effect from a plot.

4.4 Two qubits, CHSH, and no-signalling

With the standard locally tomographic complex composite, the singlet state is

$$\rho_- = \frac{1}{4} \left(I \otimes I - \sum_{j=x,y,z} \sigma_j \otimes \sigma_j \right). \quad (29)$$

Projective outcomes $a, b \in \{\pm 1\}$ along unit vectors $\mathbf{x}, \mathbf{y}$ have

$$P(a, b \mid \mathbf{x}, \mathbf{y}) = \frac{1}{4} [1 - ab \mathbf{x} \cdot \mathbf{y}], \quad (30)$$

$$E(\mathbf{x}, \mathbf{y}) = -\mathbf{x} \cdot \mathbf{y}. \quad (31)$$

The standard CHSH settings give

$$|S_{\text{CHSH}}| = 2\sqrt{2} \quad (32)$$

and $\sum_b P(a, b \mid \mathbf{x}, \mathbf{y}) = 1/2$, independent of $\mathbf{y}$. Bell locality is violated, while operational no-signalling is retained [29–31]. Loophole-free tests establish the physical severity of this constraint [32, 33].

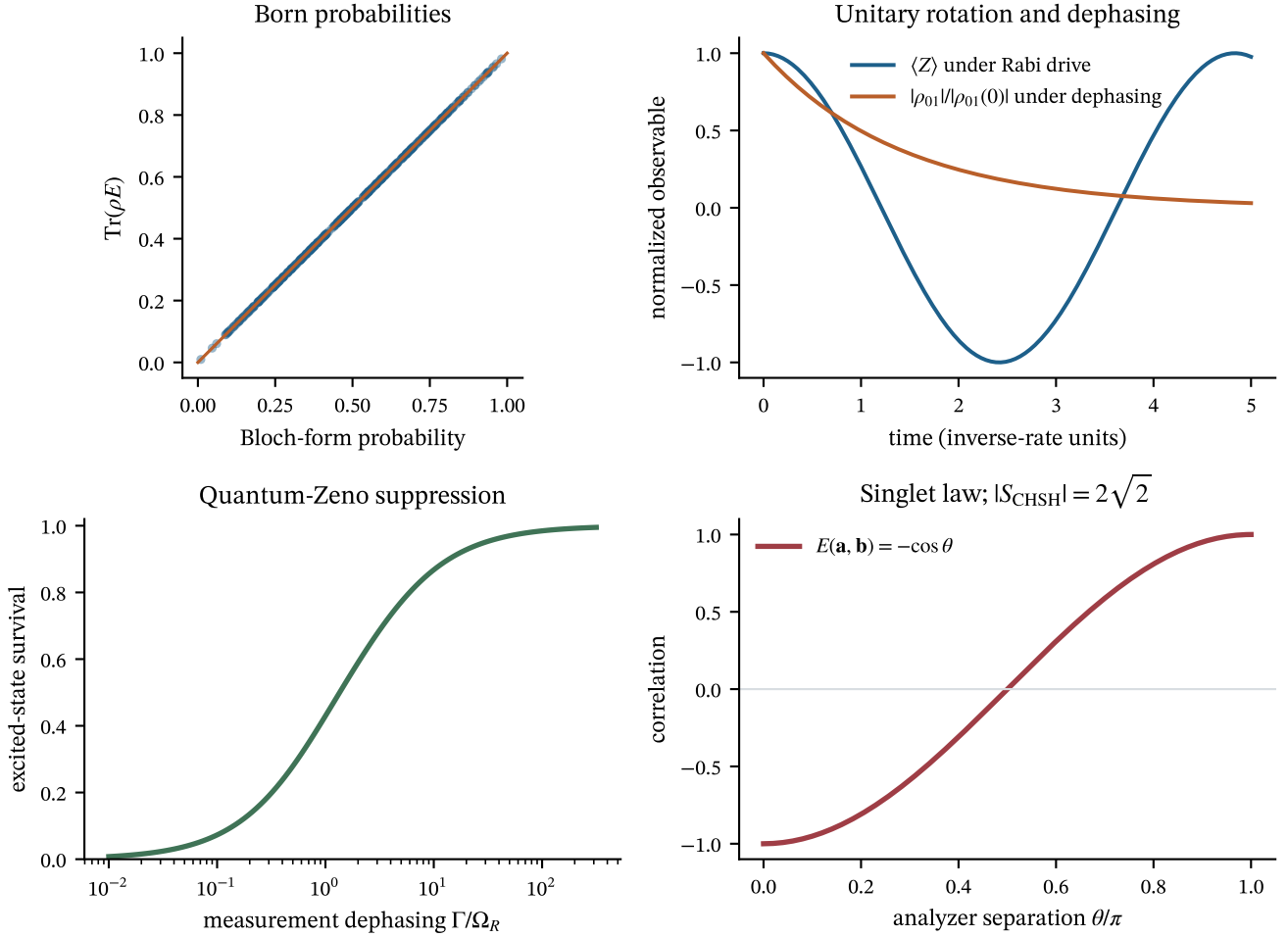


Figure 4: Internal regression checks for Court 3. The Born trace identity, unitary/dephasing laws, Zeno suppression, singlet correlation, Tsirelson value, and no-signalling marginals agree with their encoded exact values to floating-point precision.

4.5 What “recover” means

At $\varepsilon = 0$, the candidate is exactly the standard complex-qubit representation and therefore inherits all results above. This is a valid embedding, not an independent derivation of every quantum instrument from weaker real dynamics. Court 3 is closed as a *consistency and limit* court. A stronger reconstruction would need to derive the full completely positive instrument calculus and composition rule without assuming them.

5 Court 4: the first allowed departure

5.1 Why a new variable must live in a projection fibre

Suppose a correction depends only on ρ , the intervention, and calibrated environment variables. It is then a modified quantum-state law, but not evidence that Q is lossy. Court 5 demands dependence on a variable u with $Q(\rho, u) = \rho$. For pure qubits, [theorem 3.1](#) identifies the smallest connected mathematical option as the coordinate $\gamma \in U(1)$.

The physical candidate is *not* an observable absolute global phase. It is a relational memory of a closed preparation history. For a loop C , define

$$\Gamma_{\text{rel}}[C] := \oint_C \mathcal{A} \pmod{2\pi}, \quad \mathcal{A} = +i\langle\psi | d\psi\rangle, \quad (33)$$

for a periodic lift, after dynamical phase and the physical reference path are included or canceled [34, 35]. For a nonperiodic gauge one must include the endpoint Pancharatnam term. Under a periodic gauge change, the closed integral changes only by

$2\pi\mathbb{Z}$, so every function of $e^{i\Gamma_{\text{rel}}}$ is gauge invariant. The local candidate state is therefore written

$$\Lambda = (\rho, \Gamma_{\text{rel}}, \eta),$$

where η denotes all matched standard quantum reference, environment, leakage, and process variables. This is not “a wave-function with observable global phase.”

Conjecture 5.1 (Holonomy-memory postulate). There exists a physical carrier for $\Gamma_{\text{rel}}[C]$ that survives the declared delay/reset operation, while all variables represented in the frozen bounded standard-quantum process state η are matched between mirror histories within specified confidence margins. Unrepresented modes remain a limitation. Geometry alone does not imply this.

5.2 Symmetry truncation

We seek a correction $g(z, \Gamma_{\text{rel}})$ to a static Z-QND scalar record. The declared leading-order class obeys:

- (S1) **Reference covariance:** g depends only on the gauge-invariant relational holonomy.
- (S2) **Periodicity:** $g(z, \Gamma_{\text{rel}} + 2\pi) = g(z, \Gamma_{\text{rel}})$.
- (S3) **Sharp QND calibration:** $g(\pm 1, \Gamma_{\text{rel}}) = 0$.
- (S4) **Mirror oddness:** reversing the oriented preparation loop sends $\Gamma_{\text{rel}} \mapsto -\Gamma_{\text{rel}}$ and $g \mapsto -g$.
- (S5) **Finite ansatz:** retain the first Fourier harmonic and an even polynomial in z of degree at most two.

Proposition 5.2 (Minimal chosen basis element). *Within the finite ansatz (S1)–(S5), the nonzero correction is unique up to a real coefficient:*

$$g(z, \Gamma_{\text{rel}}) = \varepsilon(1 - z^2) \sin \Gamma_{\text{rel}}. \quad (34)$$

Proof. Periodicity and mirror oddness give a sine series $g = \sum_{n \geq 1} f_n(z) \sin(n\Gamma_{\text{rel}})$. Sharp calibration requires $f_n(\pm 1) = 0$. The lowest even polynomial with those zeros is proportional to $1 - z^2$. Retaining $n = 1$ gives equation (34). $\square$

This is a basis choice inside a declared truncation, not uniqueness under symmetry alone. Arbitrary smooth functions of the form

$$(1 - z^2)h(z) \sin \Gamma_{\text{rel}} \quad \text{or} \quad (1 - z^2)z^{2k} \sin(n\Gamma_{\text{rel}}),$$

as well as higher harmonics, derivative memory, and additional environment coordinates, remain symmetry-allowed. They are set to zero before synthetic target generation and define the alternative-model class.

5.3 Pure-state history-dependent record ansatz

For the target homodyne setting $\varphi = 0$, define

$$dY_t = a [z_t + \varepsilon(1 - z_t^2) \sin \Gamma_{\text{rel},t}] dt + dW_t, \quad (35)$$

$$ds_t = a dY_t - a^2 \varepsilon(1 - z_t^2) \sin \Gamma_{\text{rel},t} dt, \quad z_t = \tanh s_t, \quad (36)$$

$$d\Gamma_{\text{rel},t} = 0 \quad \text{during the probe.} \quad (37)$$

The memoryless reduced-qubit null is $\varepsilon = 0$. Applying Itô’s lemma to $z = \tanh s$ gives

$$dz_t = a(1 - z_t^2)dW_t. \quad (38)$$

Thus the declared projected population remains a bounded martingale and the eigenstates remain fixed. The added record drift does not, however, define a mixed-state, completely positive, composable measurement instrument. VD-Hopf-1 is a pure-state phenomenological observation ansatz whose full backaction, environment carrier, and multipartite completion remain open.

5.4 Known-theory warning

A physical phase reference, residual cavity field, pulse distortion, leakage level, coherent spectator mode, or non-Markovian bath can produce history-dependent readout within ordinary quantum mechanics. Quantum reference frames specifically convert apparently absolute phase into a relation in an enlarged system [19, 20]. Therefore a nonzero fitted ε against a memoryless reduced-qubit model would establish only that the reduced state was not Markov-sufficient. Court 6 requires explicitly bounded qutrit–cavity and process-tensor nulls; even their failure would not exclude an arbitrarily large quantum environment.

5.5 No-signalling constraint

A deterministic nonlinear response that distinguishes ensembles with the same density matrix can enable superluminal signalling when remote steering is available [13, 14]. Parameter independence at the ontic level is sufficient but not necessary. The required operational constraint is instead

$$\int P(o_A | a, b, \lambda) d\mu(\lambda | a, b) \text{ is independent of } b \quad (39)$$

for every local setting a , outcome o_A , and remote setting b , together with a complete multipartite intervention law. No such $\varepsilon \neq 0$ completion is supplied here. Consequently VD-Hopf-1 is a local phenomenological model, not a replacement theory.

6 Court 5: a same- ρ separating pair

6.1 Closed loops and Hopf holonomy

Let a pure qubit ray traverse a closed loop C on the Bloch sphere and return to ρ_0 . The natural spin- $\frac{1}{2}$ connection gives geometric phase

$$\Gamma_{\text{rel}}[C] = -\frac{\Omega(C)}{2} \pmod{2\pi}, \quad (40)$$

where $\Omega(C)$ is the oriented solid angle. This is the Berry phase in the adiabatic case and the Aharonov–Anandan phase for general cyclic evolution [36, 37]. Reverse orientation changes its sign.

Berry phase has already been measured in superconducting qubits using closed paths, spin echo, and interferometric and tomographic readout [38]; orientation-dependent geometric dephasing is also established within open quantum mechanics [39]. The proposed test is not “observe Berry phase again.” It deliberately restores the same system density matrix, removes the interferometric phase readout, resets or characterizes the apparatus, and asks whether a later identical probe retains orientation information beyond frozen open-system models.

Prepare mirror histories C_+ and C_- enclosing $+\Omega$ and $-\Omega$, while calibrating away dynamical phase. Their candidate deep states are

$$\Lambda_{\pm} = (\rho_0, \Gamma_{\text{rel},0} \mp \Omega/2, \eta_0). \quad (41)$$

The projected states satisfy

$$Q(\Lambda_+) = Q(\Lambda_-) = \rho_0. \quad (42)$$

For $\Gamma_{\text{rel},0} = 0$, equation (35) predicts opposite record corrections.

Writing the same η_0 in both arms invokes conjecture 5.1; it is not established by the equality of reduced density matrices. If a cavity, controller, or reference carries the orientation in ordinary quantum states $\eta_+ \neq \eta_-$, a later detector can distinguish the arms within standard quantum mechanics. If the complete joint state is truly reset to the same factorized state, standard quantum mechanics predicts equality and any ordinary carrier is erased.

Proposition 6.1 (Ordinary-quantum coherent-memory countermodel). *Let a qubit have the same reduced state ρ_0 after both histories, while a cavity retains coherent states $|\alpha\rangle$ and $|\alpha\rangle$:*

$$\rho_{SM}^{(\pm)} = \rho_0 \otimes |\pm\alpha\rangle\langle\pm\alpha|.$$

Both arms have the same cavity photon number $|\alpha|^2$, but homodyne measurement along the aligned quadrature distinguishes them. For real α , the total-variation distance is

$$D_{\text{TV}} = 2\Phi(2|\alpha|) - 1 > 0 \quad (\alpha \neq 0).$$

Proof. The aligned quadrature $X = (a + a^\dagger)/\sqrt{2}$ has distributions $\mathcal{N}(\pm\sqrt{2}\alpha, 1/2)$. The equal-variance Gaussian total-variation formula gives the result. $\square$

Therefore equality of reduced qubit states and cavity *population* is insufficient; complex cavity amplitude and every bounded modeled memory coordinate require equivalence margins.

Theorem 6.2 (Protocol prediction inside VD-Hopf-1). *For mirror loops followed by the same static probe, VD-Hopf-1 obeys*

$$\Delta_\star(\Omega) := \frac{\mathbb{E}[Y_T | C_+] - \mathbb{E}[Y_T | C_-]}{T} = -2a\varepsilon \cos \Gamma_{\text{rel},0} \sin \frac{\Omega}{2} \left[\frac{1}{T} \int_0^T \mathbb{E}(1 - z_t^2) dt \right]. \quad (43)$$

For $a^2T \ll 1$ away from the poles,

$$\Delta_\star(\Omega) = -2a\varepsilon(1 - z_0^2) \cos \Gamma_{\text{rel},0} \sin \frac{\Omega}{2} [1 + O(a^2T)]. \quad (44)$$

The memoryless reduced-qubit null predicts $\Delta_\star = 0$. A standard enlarged quantum model may predict a difference if its matched-state condition fails.

Proof. The projected z_t process is the same in both arms under [equation \(38\)](#). Insert $\Gamma_{\text{rel},\pm} = \Gamma_{\text{rel},0} \mp \Omega/2$ into [equation \(35\)](#), subtract, and use $\sin(u-v) - \sin(u+v) = -2 \cos u \sin v$. The short-time form follows from $\mathbb{E}(1 - z_t^2) = (1 - z_0^2)[1 + O(a^2t)]$. $\square$

The sine-law magnitude is 4π -periodic in solid angle and maximal at $\Omega = \pi$ modulo 2π ; the signed extremum depends on $\text{sgn}(\varepsilon \cos \Gamma_{\text{rel},0})$. Orientation oddness rejects pulse-energy effects that are even under reversal, but does not by itself reject coherent hardware memory.

This preserves the strongest methodological feature of the earlier Voss N5 proposal—compare forward and reverse histories under a frozen odd statistic—while replacing a phenomenological lag angle with a canonical bundle holonomy. Geometry supplies the candidate history label; it does not supply the nonzero readout coupling ε .

6.2 Operational defect hierarchy

Let $H \in \{+, -\}$ be the randomized history label. The directly estimable reduced-state witness is

$$\mathcal{E}_{\text{sys}} := I(H; Y_{0:T} \mid \rho_0, A_{\text{QND}}). \quad (45)$$

Under the short Gaussian approximation,

$$\tilde{Y}_T := Y_T - az_0T, \quad \tilde{Y}_T \mid H = \pm \sim \mathcal{N}(\mp m, T), \quad m = a\varepsilon(1 - z_0^2)T \cos \Gamma_{\text{rel},0} \sin(\Omega/2),$$

so $\mathcal{E}_{\text{sys}} > 0$ whenever $m \neq 0$.

6.3 Direct fibre-variation optimization

Fix a bounded probe protocol $p = (a, T, \text{instrument family})$ and fix all declared standard quantum process variables to the matched value η_0 . Define the candidate slice

$$\mathcal{F}_{\rho, \eta_0} := \{(\rho, \Gamma_{\text{rel}}, \eta_0) : \Gamma_{\text{rel}} \in S^1\}.$$

This restriction is essential: allowing arbitrary environments inside the fibre would make history dependence routine in open quantum mechanics. The protocol-level Court-5 objective is

$$V_p(\rho; \eta_0) := \sup_{\Lambda_1, \Lambda_2 \in \mathcal{F}_{\rho, \eta_0}} D_{\text{TV}}(P_p(\cdot \mid \Lambda_1), P_p(\cdot \mid \Lambda_2)), \quad (46)$$

$$V_{p, \eta_0}^\star := \sup_{\rho \in S_{\text{pure}}} V_p(\rho; \eta_0). \quad (47)$$

Then $V_{p, \eta_0}^\star > 0$ witnesses insufficiency of the matched reduced-state/protocol model. It is not a supremum over unbounded measurement strength or time; that unrestricted supremum would approach one for any nonzero signal.

For VD-Hopf-1 in the short Gaussian probe, optimizing over $\sin \Gamma_{\text{rel},1} = +1$ and $\sin \Gamma_{\text{rel},2} = -1$ gives

$$V_p(\rho; \eta_0) \simeq 2\Phi(a|\varepsilon|(1 - z^2)\sqrt{T}) - 1, \quad V_{p, \eta_0}^\star \simeq 2\Phi(a|\varepsilon|\sqrt{T}) - 1. \quad (48)$$

Thus this protocol-fixed candidate approximation has positive variation if and only if $\varepsilon \neq 0$; its maximum occurs at the equator. This is a computation inside the candidate law. Deriving a nonzero ε from the reversible Ψ , or measuring one in Nature, remains open.

Search failure is not a no-go theorem. A numerical optimizer returning zero on a finite ansatz establishes only failure of that search. Global closure requires proof over a declared complete adaptive control and measurement class. The relevant sufficient criterion is factorization.

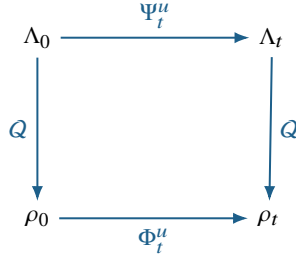
Proposition 6.3 (Court-5 factorization criterion). *Suppose that for every control history u there is a projected dynamics Φ_t^u , and for every adaptive control/instrument history $\mathbf{A}$ there is a projected process law $\tilde{P}$, such that*

$$Q(\Psi_t^u(\Lambda)) = \Phi_t^u(Q(\Lambda)), \quad (49)$$

$$P(\mathbf{O} \mid \mathbf{A}, \Lambda, \eta_0) = \tilde{P}(\mathbf{O} \mid \mathbf{A}, Q(\Lambda), \eta_0). \quad (50)$$

Then every protocol-level matched fibre variation is zero.

Proof. If $\Lambda_1, \Lambda_2 \in \mathcal{F}_{\rho, \eta_0}$, equation (49) keeps their projected controlled histories equal and equation (50) makes every allowed adaptive outcome law equal. Every total-variation distance in equation (46) is therefore zero. This proposition states the exact closure criterion; the open problem is to derive its premises from the candidate architecture rather than assume the conclusion. $\square$



6.4 What failure would and would not mean

Court 5 attacks *predictive completeness*, not every possible modification of the accepted dynamics:

Representational completeness: is Q one-to-one?

Predictive completeness: if Q is many-to-one, is every matched fibre operationally gauge for every bounded allowed protocol?

Dynamical correctness: even if ρ is sufficient, does Nature obey the accepted quantum law for ρ ?

A global Court-5 no-go would remove the claim that ρ discards observable deeper information. It would not rule out a modified law

$$\dot{\rho} = \mathcal{L}_{\text{QM}}(\rho) + \varepsilon \mathcal{F}(\rho, C)$$

with a one-to-one deeper representation. Nor would a null in weak Markovian circuit QED prove global factorization under strong memory, measurement reversal, rapid reference modulation, temporally ordered probes, or composition.

Three evidentiary levels must not be conflated:

$\mathcal{E}_{\text{sys}} > 0$: the reduced Markov qubit state is insufficient.

Bounded process-model excess: a preregistered completely positive multi-time comb/process tensor with frozen dimension and memory capacity fails on held-out loops [40–44]. Initial correlations and reduced-dynamics domain assumptions remain relevant [45].

Beyond-quantum claim: all physically admissible quantum enlargements, including reference and environment memory, are excluded. Finite laboratory characterization cannot literally enumerate an unbounded environment; such a claim requires bounded-resource assumptions or device-independent inequalities.

This hierarchy prevents a memory effect from being mislabeled as a failure of quantum mechanics.

6.5 Outcome ladder and search order

Level	Mathematical result	Scientific status
A	Derive $\rho = Q(\xi, R, E)$ and recover quantum predictions.	Explanatory reconstruction.
B	Prove Q is many-to-one.	Representational redundancy; not new physics.
C	Establish matched, protocol-bounded $V^* > 0$.	The declared reduced quantum state/process model is predictively incomplete.
D	If C fails, establish $P_{\text{alt}}(\mathbf{O} \mid \rho, a) \neq P_{\text{accepted}}(\mathbf{O} \mid \rho, a)$.	State sufficient; a modified-dynamics candidate is required and must be tested separately.
E	Both C and D fail over the tested domain.	The declared quantum models remain adequate there; Voss Dynamics is explanatory/methodological.

The search order is

single system $\longrightarrow$ temporal process $\longrightarrow$ system+reference $\longrightarrow$ composites $\longrightarrow$ Bell/network scenarios.

Single-system factorization does not imply composite factorization. A shared reference may support a relational variable $C(\Lambda_A, \Lambda_B, R)$ that is absent from the product of local reduced states. Conversely, any composite proposal must satisfy the no-signalling and Bell constraints already stated. Only a no-go theorem across the declared hierarchy closes the strong predictive-incompleteness branch.

7 Court 6: freeze, measure, and score

7.1 Minimal apparatus

The target needs one high-coherence qubit, phase-coherent control capable of a closed ray loop, and a nominally Z-QND readout. A superconducting transmon is natural because dispersive readout, quadrature backaction, trajectory validation, and non-Markovian characterization are mature [22, 23, 26, 44]; it is difficult because strong dispersive readout also causes state mixing and level-dependent effects [46]. The experiment has five temporal blocks:

1. prepare the same equatorial state ρ_0 ;
2. randomly assign C_+ , C_- , or a sham loop;
3. close the ray loop and verify final-state equality on a tomography subset;
4. actively reset or characterize the cavity/reference state;
5. acquire a short nominally Z-QND record with no Rabi drive.

7.2 Draft calibration and synthetic target

Independently calibrate a , z_0 , readout noise, loop solid angle, the reference offset $\Gamma_{\text{rel},0}$, dynamical phase cancellation, leakage, cavity ringdown, and control transfer functions. The illustrative software test fixes $\Gamma_{\text{rel},0} = 0$. Use only a designated calibration set of solid angles $\{\Omega_j^{\text{cal}}\}$ to estimate ε :

$$\widehat{\varepsilon} = \arg \min_{\varepsilon} \sum_{j \in \text{cal}} \frac{[\widehat{\Delta}_j + 2a\varepsilon(1 - z_0^2) \sin(\Omega_j/2)]^2}{\sigma_j^2}. \quad (51)$$

Then freeze

$$\Delta_j^{\text{frozen}} = -2a\widehat{\varepsilon}(1 - z_0^2) \sin(\Omega_j^{\text{hold}}/2) \quad (52)$$

before revealing any held-out record. The repository contains a machine-readable *synthetic* freeze specification and a reproducibility manifest. Regenerating that manifest is not an external timestamped commitment; a physical study requires an independent registry or cryptographic timestamp before acquisition.

7.3 Likelihood and baselines

The intended primary score is held-out path likelihood with no refitting. The current code implements only a cell-level integrated-current Gaussian summary score with propagated calibration covariance; it is a software smoke test, not the final path analysis. A confirmatory implementation must compare:

- (M1) **Memoryless reduced-qubit null:** $\varepsilon = 0$.
- (M2) **Bounded calibrated quantum process tensor:** completely positive multi-time model with frozen time grid, intervention basis, memory length/bond dimension, regularization, and confidence set.
- (M3) **VD-Hopf-1:** the one-parameter frozen sine law.
- (M4) **Artifact baselines:** pulse energy, duration, signed amplitude leakage, cavity residue, IQ imbalance, thermal drift, and flexible low-rank history models.

Model selection reports predictive log likelihood and complexity separately. Minimum description length is a model-selection rule, not a collapse mechanism. The capacity of every flexible history model must be bounded before target reveal; an unrestricted history model can reproduce the sine law exactly.

7.4 Power

In the short-probe approximation, a single rate estimate Y_T/T has variance $1/T$. With N records per orientation,

$$\text{SE}(\hat{\Delta}) = \sqrt{\frac{2}{NT}}.$$

For one-sided threshold z_α and power $1 - \beta$,

$$N \geq \frac{2(z_\alpha + z_{1-\beta})^2}{4a^2\varepsilon^2(1 - z_0^2)^2T \cos^2 \Gamma_{\text{rel},0} \sin^2(\Omega/2)}. \quad (53)$$

At 5σ , 90% oracle power, $a = 1$, $z_0 = 0$, $\Gamma_{\text{rel},0} = 0$, $T = 0.05$, $\Omega = \pi$, and $|\varepsilon| = 0.02$, the independent-white-noise approximation requires roughly 9.86×10^5 records per orientation. This excludes calibration uncertainty, nuisance estimation, autocorrelation, drift blocks, multiplicity, exclusions, tomography, and process-model comparison; full simulation-based power will be larger.

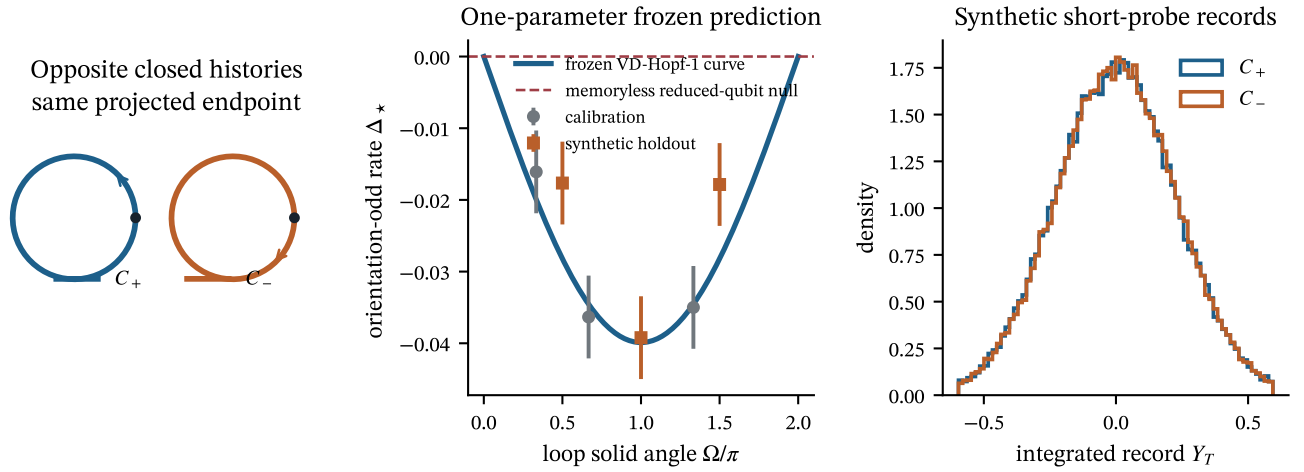


Figure 5: The tiny target. Closed histories have the same projected endpoint by construction, the one-parameter ansatz freezes a solid-angle sine curve, and synthetic summaries exercise the regression code. The points are synthetic and do not validate a physical memory carrier, complete instrument, or process-tensor comparison.

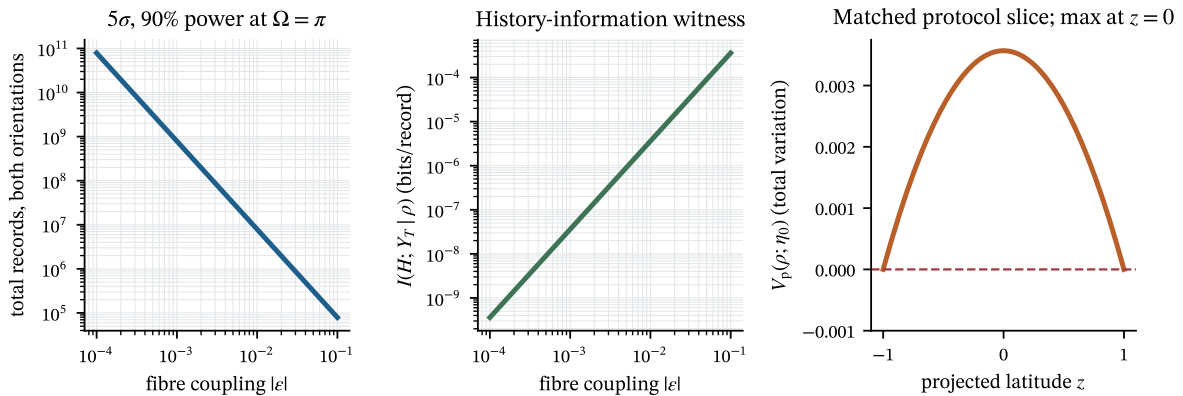


Figure 6: Left: oracle record cost for 5σ and 90% power. Center: binary-history mutual information per record. Right: the matched protocol-level variation $V_p(\rho; \eta_0)$, maximized at the equator. All panels are calculations inside the short Gaussian ansatz.

7.5 Controls

The following controls are mandatory:

- **Final-state equivalence:** a one-sided equivalence test using SPAM-robust tomography or gate-set tomography bounds the trace distance tightly enough that the largest permitted $z_+ - z_-$ cannot explain the target. At the illustrative $a = 1, \varepsilon = 0.02, \Omega = \pi$ point, a 0.04 population mismatch reproduces the entire leading rate difference. Randomized benchmarking is auxiliary and cannot establish state equality.
- **Leakage:** characterize at least the qutrit populations and level-dependent IQ response; equality of normalized 2×2 blocks is insufficient.
- **Dynamical phase:** freeze explicit $H_{\pm}(t)$, calculate the dynamical phase, reconstruct the realized state path, and set a numerical residual-phase margin. “Use echo” is not a complete protocol.
- **Ringdown:** measure complex cavity amplitude and covariance, not photon number alone; vary delay and fit a frozen multi-mode ringdown model.
- **Sham and zero-area loops:** equal pulse count, energy, and duration with $\Omega = 0$ must be null.
- **Orientation swaps:** cable/IQ labels, pulse compilation, and readout sign are independently reversed.
- **Loop families:** geometrically distinct paths with the same Ω must agree; equal-duration paths with different Ω must follow the sine law.
- **Process tensor:** freeze time grid, intervention basis, leakage dimension, memory length/bond dimension, regularization, stationarity assumptions, confidence set, and out-of-distribution rule.
- **Replication:** repeat with a second control implementation and independent analysis team.

7.6 Hard pass/fail rule

The draft is not executable until it has: a mixed-state or bias-qualified pure-state extension; a minimum composable instrument-consistency result; explicit control Hamiltonians and reset law; implemented path score; validated bounded qutrit–cavity and process-tensor nulls; numeric thresholds, sample count, autocorrelation correction, missing-data rules, and equivalence margins; frozen model capacities; and an external commitment. Only after those blockers are closed can VD-Hopf-1 enter a bounded-model laboratory stage, where:

1. the frozen held-out likelihood exceeds the memoryless reduced-qubit, qutrit–cavity, and declared finite-memory models by the preregistered threshold;
2. the full solid-angle, orientation, z_0 , delay, and reference-phase dependence is jointly correct;
3. no post-unsealing parameter change is made;
4. every equivalence and negative-control margin holds; and
5. the bounded calibrated process-tensor prediction fails outside its frozen confidence set.

Independent replication is a separate confirmation stage. Even a single-stage pass establishes unexplained history dependence relative to the listed bounded models, not exclusion of all enlarged quantum models. If $\hat{\varepsilon}$ is consistent with zero, VD-Hopf-1 is bounded or rejected at the achieved scale; the conditional scalar characterization and QND coordinate method may remain.

8 Computational evidence

8.1 Reproducible suite

The evidence package performs:

1. symbolic verification of both Itô residuals for the QND invariant and quotient;
2. the integer scalar-selection calculation $2 = 1 + d(d - 1)/2$;
3. positivity, trace, and global-phase invariance tests for Q ;
4. Born trace identities on random Bloch states;
5. unitary invariants, dephasing, Rabi, and Zeno calculations;
6. singlet correlations, $2\sqrt{2}$ CHSH, and no-signalling marginals;
7. deformation symmetry tests;
8. algebraic invariance of $\psi\psi^\dagger$ under inserted opposite phases (not a loop/control simulation);
9. an ordinary-quantum coherent-memory countermodel with equal photon number and nonzero homodyne distinguishability;
10. direct evaluation of $V_p(\rho; \eta_0)$ and V_{p, η_0}^* in the short-probe candidate;
11. frozen synthetic calibration/holdout scoring, power, and mutual information.

The automated test suite passes all unit tests. Representative maximum errors are 2.3×10^{-16} for Born probabilities, 7.8×10^{-16} for Rabi evolution, 1.2×10^{-16} for no-signalling marginals, and below 5×10^{-16} for global-phase projection. These are floating-point checks of exact identities.

8.2 What the synthetic freeze proves

Synthetic Gaussian summaries generated from VD-Hopf-1 show that the calibration–freeze–reveal regression can recover a small coupling and predict held-out angle cells without refitting. They do not validate path likelihood, a complete instrument, or process-tensor discrimination. Because the simulator contains the fitted functional family, this is a software/protocol smoke test only.

9 Impact if the program succeeds

9.1 A physical origin for complex structure

[Theorem 2.7](#) offers a conditional dimension criterion for a familiar mathematical fact. A real stationary reference produces a quarter-turn operator $J_R^2 = -I$; its two-dimensional span matches the complete unnormalized binary QND-filter algebra only at Peirce degree two. If the accessibility and response-completeness premises can be derived from weaker coupling principles, the criterion could contribute to explaining why complex amplitudes are selected.

9.2 A sharper meaning of “emergent”

The map Q is not a decoder metaphor. It is an explicit quotient:

$$\text{real normalized spinor and reference} \longrightarrow \text{complex ray/density matrix} \longrightarrow \text{operational predictions.}$$

The fibre distinction identifies what the redundant pure-spinor lift retains and what ρ identifies. Standard quantum mechanics declares the global $U(1)$ direction gauge. VD-Hopf-1 postulates a distinct closed-history carrier and asks whether a residual relation predicts a later record after a bounded matched reset. A positive result would first establish insufficiency of the declared reduced-state/process model, not immediate failure of all quantum theory.

9.3 Measurement and randomness

The QND chart shows that a real measurement record can drive both information gain and phase backaction along one trajectory while leaving a transverse state label conserved. This is standard quantum filtering, but the chart makes the quotient geometry constructive. If a fibre coupling existed, some apparent measurement randomness would become conditionally structured by Γ_{rel} . That would not make outcomes globally deterministic without a complete environment law; it would show only that ρ omitted predictive information.

9.4 Quantum engineering

Even a null result can be useful. The protocol measures how well a reduced qubit state or calibrated process tensor predicts records after closed loops. It therefore probes coherent pulse memory, cavity reset, geometric control, and low-rank non-Markovian structure. The supplied observability search also yields a principled single-meter control design, though continuous one-axis tomography with Rabi control is established [47]. The engineering contribution is disciplined model discrimination, not a new detector.

9.5 Scientific method

The most transferable contribution is the chain

$$\boxed{\text{derive} \longrightarrow \text{commit} \longrightarrow \text{measure} \longrightarrow \text{score} \longrightarrow \text{replicate.}}$$

Foundational claims are unusually vulnerable to flexible reinterpretation. A symmetry-truncated deformation, externally committed target, bounded adversarial quantum-memory null, machine-readable constants, and hard loss condition can convert an interpretive proposal into an experiment that can say “no.” The present repository prepares that workflow but is not itself an externally registered physical study.

10 Limitations and open theorems

1. **The EJA premise is not derived.** Homogeneity, self-duality, finite dimensionality, and irreducibility delimit the theorem.

2. **Generator completeness is an axiom.** It may be physically motivated, but it currently performs the decisive scalar selection.
3. **The reference-to-state relation is not unique.** $J_R^2 = -I$ gives a complex structure; identifying the same reference with every system phase requires a coupling law.
4. **The holonomy carrier is postulated.** A path functional is not automatically an endpoint state coordinate; retaining it while matching the full standard quantum process state is the decisive conjecture.
5. **The Hopf coupling is conjectural.** The sine term is one selected finite-basis element, not a symmetry-unique law.
6. **Global phase is ordinarily gauge.** A physical reference enlarges the quantum system and can turn apparent global phase into standard relative phase. The experiment must beat the declared bounded enlarged-quantum null.
7. **The ansatz is pure-state and noncomposable.** No canonical mixed-state holonomy, completely positive instrument, qutrit extension, or adaptive multipartite law is derived.
8. **A local model is not a composite theory.** No-signalling, Bell correlations, and preparation contextuality must coexist in a complete multipartite extension.
9. **Finite process tomography is bounded.** Failure of a chosen process-tensor model is not logical exclusion of every quantum environment.
10. **The protocol is not yet executable.** Numeric decision thresholds, capacity bounds, path likelihood, qutrit–cavity simulations, equivalence margins, and an external commitment remain.
11. **No laboratory result is included.** The thesis reaches a conditional-theorem and candidate-protocol milestone, not physical confirmation.

11 Conclusion

The six courts force the broad idea of Emergent Predictive Representation into a small and auditable statement.

The conditional reconstruction begins with operational convexity, a Jordan-domain assumption, strengthened measure-preserving or Hamiltonian dynamics, and a stationary real detector reference. The reference span has two real directions. A binary Jordan face has $1 + d(d - 1)/2$ unnormalized QND filter directions. The dimension-forcing response axiom selects $d = 2$, conditionally recovering complex quantum scalars.

The effective pure-qubit map can be represented by the redundant Hopf lift $\psi \mapsto \psi\psi^\dagger$. The ideal QND equations become a straight translation in (s, ϑ) , but their conserved transverse coordinate remains encoded in ρ . The chosen normalized-spinor lift has a $U(1)$ fibre; promoting a closed-history relation on that fibre to physical memory is an explicit candidate postulate, not a consequence of quantum-state geometry.

The leading orientation-odd candidate readout correction is

$$dY_t = a[z_t + \varepsilon(1 - z_t^2) \sin \Gamma_{\text{rel},t}]dt + dW_t,$$

and its frozen small-target signature is

$$\Delta_\star(\Omega) \simeq -2a\varepsilon(1 - z_0^2) \cos \Gamma_{\text{rel},0} \sin(\Omega/2)$$

Equivalently, the direct candidate optimization gives

$$V_{p,\eta_0}^\star \simeq 2\Phi(a|\varepsilon|\sqrt{T}) - 1.$$

The repository specifies a draft calibration, holdout, Gaussian summary score, power calculation, controls, hashes, and qualitative failure rules. It does not yet implement the bounded process-tensor and qutrit–cavity nulls needed before a confirmatory run. If a genuine adaptive factorization theorem instead forces all matched protocol variations to zero, the claim that ρ discards observable deeper information is closed cleanly; the conditional complex-scalar characterization survives, and a logically separate modified- ρ dynamics branch remains available. The scientifically credible conclusion today is that a one-parameter phenomenological target and explicit theory/protocol closure conditions have been isolated for further work.

A Full matrix proof of the unnormalized QND-filter stabilizer

Use coordinates $v = (t, z, c)$ and Lorentz metric $\eta = \text{diag}(1, -1, -I_d)$. Modulo a scalar dilation, an infinitesimal cone automorphism X obeys

$$X^\top \eta + \eta X = 0.$$

Write

$$X = \begin{pmatrix} a & b & \mathbf{u}^\top \\ c & d & \mathbf{v}^\top \\ \mathbf{p} & \mathbf{q} & M \end{pmatrix}.$$

The Lorentz condition gives $a = d = 0$, $b = c$, $\mathbf{p} = \mathbf{u}$, $\mathbf{q} = -\mathbf{v}$, and $M^\top = -M$. Preserving each ray generated by $n_\pm = (1, \pm 1, \mathbf{0})$ requires

$$Xn_\pm \in \text{span}(n_\pm).$$

The transverse components are $\mathbf{p} \pm \mathbf{q}$, so both vanish and hence $\mathbf{p} = \mathbf{q} = \mathbf{u} = \mathbf{v} = \mathbf{0}$. The remaining matrix is

$$X = b \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & M \end{pmatrix}, \quad M \in \mathfrak{so}(d),$$

which proves [equation \(10\)](#). Stabilizing only one null ray would leave null rotations; stabilizing both is essential.

B Itô derivation of the relational QND chart

For $s(z) = \text{artanh } z$,

$$s'(z) = \frac{1}{1-z^2}, \quad \frac{1}{2}s''(z) = \frac{z}{(1-z^2)^2}.$$

Using [equation \(19\)](#),

$$(dz)^2 = a^2 \cos^2 \varphi (1-z^2)^2 dt.$$

Therefore

$$\begin{aligned} ds &= s'(z)dz + \frac{1}{2}s''(z)(dz)^2 \\ &= a \cos \varphi dW + a^2 \cos^2 \varphi z dt \\ &= a \cos \varphi dY. \end{aligned}$$

Substitution of [equation \(18\)](#) into [equation \(20\)](#) likewise gives $d\vartheta = a \sin \varphi dY$. The orthogonal rotation to (x_φ, c_φ) then yields [equation \(24\)](#). The symbolic evidence suite differentiates the candidate first integral directly and verifies both its diffusion and drift residuals are identically zero.

C Why exact QND apparatus readout is phase blind

An exact Z-QND system–environment unitary has controlled form

$$U = |0\rangle\langle 0| \otimes U_0 + |1\rangle\langle 1| \otimes U_1.$$

For $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ and an initially uncorrelated, preparation-phase-independent environment state $|\psi\rangle\langle\psi| \otimes \rho_E$, tracing out the qubit after interaction gives

$$\rho'_E = |\alpha|^2 U_0 \rho_E U_0^\dagger + |\beta|^2 U_1 \rho_E U_1^\dagger.$$

The cross terms vanish because $\text{Tr}_S(|0\rangle\langle 1|) = 0$. Hence every environment-only POVM has probabilities independent of $\arg(\alpha^* \beta)$. No nonlinear statistic of that exact apparatus-only record can recover the initial relative phase. A drive, non-QND term, unrecovered cavity mode, or joint system–environment recombination can reveal phase within standard quantum mechanics; those are different experiments.

D Deformation expansion beyond leading order

The general smooth reference-covariant, mirror-odd correction is

$$g(z, \Gamma_{\text{rel}}, \mathbf{h}) = \sum_{n \geq 1} f_n(z, \mathbf{h}) \sin(n\Gamma_{\text{rel}}),$$

where h denotes causal history features. Sharp eigenstate calibration imposes $f_n(\pm 1, h) = 0$. For an analytic static model,

$$f_n(z) = (1 - z^2) \sum_{k \geq 0} (a_{nk} z^{2k} + b_{nk} z^{2k+1}).$$

Additional branch-relabel and time-reversal conventions determine whether the a - or b -sector survives. VD-Hopf-1 keeps $a_{10} = \varepsilon$ and sets all other coefficients to zero before target data. An experimental rejection can therefore mean either $\varepsilon = 0$ or that this truncation is wrong; it does not reject every conceivable history-dependent model.

E Likelihood, information, and power

For equal priors and the short Gaussian pair

$$Y \mid H = \pm \sim \mathcal{N}(\pm m, T),$$

the log likelihood ratio is $2mY/T$. The mutual information in bits is

$$I(H; Y) = 1 - \mathbb{E}_{Z \sim \mathcal{N}(0,1)} \log_2 \left[1 + \exp \left(-\frac{2m(m + \sqrt{T}Z)}{T} \right) \right].$$

The evidence code evaluates this integral by Gauss–Hermite quadrature. At small m^2/T , information is quadratic in ε , matching the inverse-square sample cost in [equation \(53\)](#).

For held-out integrated records y_i with known variances T_i , the primary Gaussian score is

$$\ell(M) = -\frac{1}{2} \sum_i \left[\frac{(y_i - \mu_i^M)^2}{T_i} + \log(2\pi T_i) \right].$$

All preprocessing, exclusions, covariance corrections, and stopping rules must be frozen with the model.

F Reproducibility map

Artifact	Purpose
evidence/epr_core.py	Relational map, Hopf projection, Jordan dimension, quantum limits, target law, power, and simulation.
evidence/run_all.py	Deterministic end-to-end evidence generation and all thesis figures.
evidence/tests/test_epr_core.py	Unit tests for positivity, fibre invariance, QND chart, Born/CHSH/no-signalling, deformation symmetries, and shot scaling.
evidence/outputs/metrics.json	Machine-readable numerical results and evidence status.
evidence/outputs/frozen_prediction.json	Candidate version, formula, calibration constants, held-out angles, and primary score.
evidence/outputs/manifest.sha256	Hashes of code and generated evidence artifacts.
blind_qnd_research.py	Supplied independent symbolic and observability-discovery seed calculation.

From the repository root:

```
uv sync --all-groups
uv run pytest
uv run python -m evidence.run_all
tectonic -X compile thesis/main.tex --outdir thesis/build
```

The random seed is fixed at 20260821. Generated target records are explicitly marked synthetic in every machine-readable output.

G Court-by-court falsification matrix

Court	Pass object	Failure condition
1	Correct conditional classification with explicit axioms.	Counterexample to the reference theorem, stabilizer lemma, or EJA classification step; hidden dimensional assumption.
2	Positive normalized Q , exact chart, correct fibres.	A claimed lost variable is reconstructible from ρ , or reference structure is merely renamed complex notation.
3	Exact $\varepsilon = 0$ quantum laws and composites.	Failure of positivity, Born calibration, CP baseline, CHSH, or no-signalling.
4	Frozen symmetry basis and remainder class.	A lower-order allowed term was omitted, or target term violates a declared symmetry.
5	Same ρ , different Λ , nonzero candidate separation.	Preparation arms end in distinguishable ρ , or predicted difference vanishes identically.
6	Sealed held-out likelihood and independent replication.	Post-hoc fitting, process-memory explanation, control failure, null result, or non-replication.

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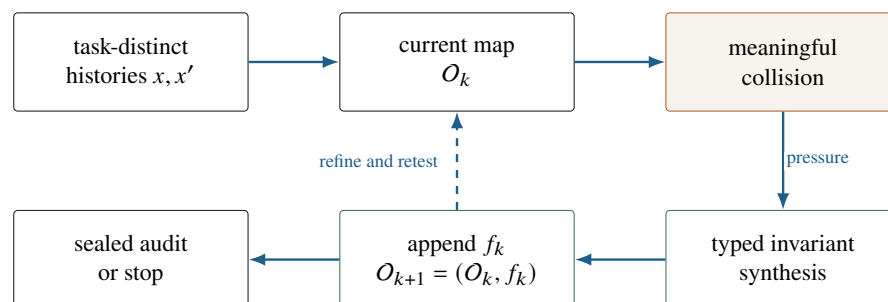
Pressure-Driven Invariant Synthesis

Collision-guided observability refinement across dynamical systems

Logan Voss

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23 August 2026



Evidence statement. The partition-refinement, finite-sample completion, threshold-nesting, and similarity-invariance results are proved under explicit assumptions. The reported numerical values are deterministic computations or internally frozen holdout audits. Three independently seeded synthetic domains show collision-risk reductions; two real UCR test sets do not establish significant transfer. The study is not externally preregistered, does not prove population injectivity, and does not claim discovery of a physical conservation law.

Abstract

Papers I and II hold the representation fixed: the first asks when dynamics and observation preserve distinctions, and the second asks whether a distinction hidden inside one representation fibre can matter for a future prediction. This paper makes the representation itself dynamic. A *meaningful collision* occurs when two task-distinct trajectories remain close under the current observation map. Such collisions become counterexamples that pressure a constrained search to synthesize a unary observable and append it to the map.

The mathematical core is simple but consequential. Exact append-only augmentation refines observation fibres; on a finite sample, a separator-complete search reaches injectivity in at most the number of unresolved class splits; and a fixed max-product metric makes threshold-collision sets nested. For trajectories transformed by translation, positive global scale, and orthogonal channel action, global RMS canonicalization and quantile recurrence preserve adjacency exactly outside deterministic tie pathologies. The resulting primitive library combines temporal, covariance, recurrence-network, and persistent-homology summaries. A depth-two typed grammar produces 271 auditable scalar programs.

The executable audit uses logistic, Lorenz, and Kuramoto systems. Programs are selected on two source domains, frozen before independent audit trajectories are materialized, and evaluated on 120 disjoint cross-label pairs from the held-out system. Collision risk falls from 0.992 to 0 for logistic, 0.917 to 0.467 for Lorenz, and 1 to 0.083 for Kuramoto; the macro absolute reduction is 0.786. A full-pipeline 99-permutation source-label null selects no program in every fold. An explicitly post-hoc phase audit links the $N = 5$ Kuramoto winner to latent second-harmonic coherence ($\rho_S = 0.907$; 0.944 within weak coupling) without upgrading the held-out claim. Yet an admissible random-program comparator is beaten significantly only for Kuramoto, and the all-source expression fails to transfer significantly to ECGFiveDays or Earthquakes. The result is therefore a verified mechanism for constrained representation repair in controlled domains, not a universal invariant-discovery claim.

Primary result Meaningful representation collisions can be converted into counterexamples for a typed, nuisance-invariant synthesis loop. In the released implementation, one scalar program is actually reinserted into the observation map and substantially reduces collision risk across three independent synthetic audits. The exact refinement guarantees are structural; empirical transfer beyond those controlled systems remains open.

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1 The third move: make representation dynamic

1.1 One problem, three directions

Voss Dynamics studies what happens to information-bearing distinctions under physical evolution and representation. Paper I asks whether a distinction between possible states or histories survives a dynamical map and a fixed observation map. Paper II asks whether two histories merged by that map can still imply different future observations. This paper begins after a predictive witness has made that failure operational: if the current map is insufficient, how should it change?

The proposed answer is not to increase dimension blindly. It is to use the failed fibre as a counterexample, search a predeclared library for a compact coordinate that separates it, reject coordinates that respond to declared nuisance transformations, append the winner, and retest. “Pressure” names this counterexample signal. It is not thermodynamic pressure. “Invariant” means invariance to a declared nuisance group. It does not mean that the synthesized observable is conserved along the physical flow.

The relation to observability is direct. Classical nonlinear observability asks whether a state can be reconstructed from an output history [14]; delay embedding supplies generic reconstruction results under smooth noiseless hypotheses [23, 27]. Here the object is narrower and more operational: a finite, task-indexed representation is repaired when it merges distinctions that a declared witness says matter.

1.2 Claim discipline

Four statuses are used throughout:

- PROVED:** A mathematical consequence of explicitly stated assumptions.
- COMPUTED:** A deterministic symbolic or numerical check reproduced by the evidence package.
- HELD-OUT:** A result on data not used by the final in-run selection step. This is internal freezing, not external preregistration or independent replication.
- OPEN:** A claim requiring new domains, prospective commitment, physical data, or stronger theory.

The complete claim ledger is supplied beside the manuscript. The release also records an earlier development failure: a first version admitted programs under permuted source labels. Gates were then strengthened, final audit seeds were changed, and the final run was executed. That history makes the final audit useful but not prospective confirmation.

1.3 The six gates

Table 1: The six gates of Paper III. Passing an earlier gate does not imply passage of a later one.

Gate	Requirement	Present result
1	Define task distinction, collision geometry, nuisance group, and claim scope.	PROVED/DECLARED: formal objects fixed in section 2 .
2	Show that augmentation truly refines the representation.	PROVED: exact refinement, finite completion, and threshold nesting.
3	Construct interpretable unary coordinates with auditable invariance.	PROVED/COMPUTED: similarity group, recurrence theorem, 24/24 adjacency checks.
4	Search, select, and actually reinsert a coordinate under fixed gates.	COMPUTED: 271-program grammar; explicit append-only implementation.
5	Survive independent controlled audits and complete-pipeline nulls.	HELD-OUT: three synthetic domains; no selection in 0/99 source-label permutations per fold.
6	Transfer without adaptation to real data.	OPEN: neither real UCR stress test is significant.

2 Meaningful collisions and representation pressure

2.1 Domains, distinctions, and fibres

Let $d \in \mathcal{D}$ index a domain, $x \in \mathcal{X}_d$ a finite trajectory, and

$$\mathcal{O}_k : \mathcal{X}_d \longrightarrow \mathbb{R}^{m_k} \quad (1)$$

the representation after k successful additions. Exact equality under $\mathcal{O}_k$ induces the fibre relation

$$x \sim_k x' \iff \mathcal{O}_k(x) = \mathcal{O}_k(x'). \quad (2)$$

A task witness declares whether their difference matters:

$$D_d(x, x') = \mathbf{1}\{\ell_d(x, x') > \delta_d\}. \quad (3)$$

The witness may encode a latent regime, a known parameter interval, a future-outcome difference, or a label mismatch. Without such a witness, every discrepancy due to measurement noise would count as a distinction, and “collision” would be scientifically underdetermined.

For the executable binary benchmarks, $D_d(x, x') = 1$ exactly when labels differ. Each coordinate is passed through the fixed source-independent squash

$$h(z) = \frac{2}{\pi} \arctan z, \quad (4)$$

and the representation distance is

$$d_k(x, x') = \max_{1 \leq j \leq m_k} |h(\mathcal{O}_{k,j}(x)) - h(\mathcal{O}_{k,j}(x'))|. \quad (5)$$

At a fixed threshold τ , define the meaningful-collision indicator and its conditional risk:

$$C_{k,\tau}(x, x') = \mathbf{1}\{D_d(x, x') = 1, d_k(x, x') \leq \tau\}, \quad (6)$$

$$R_{d,k}(\tau) = \mathbb{P}_d(d_k(X, X') \leq \tau \mid D_d(X, X') = 1). \quad (7)$$

The benchmark fixes $\tau = 0.22$. Absolute reduction,

$$\Delta R_d(f) = R_{d,k}(\tau) - R_{d,k+1}(\tau), \quad (8)$$

is primary. Relative reduction is unstable when the baseline risk is small.

2.2 Pressure is a counterexample, not a metaphorical score

A sampled pair with $C_{k,\tau} = 1$ is a concrete counterexample to sufficiency at resolution τ . It says that the current coordinates cannot separate this task-distinct pair. A candidate must be a *unary* map $f : \mathcal{X}_d \rightarrow \mathbb{R}$: pairwise diagnostics such as $g(x - x')$ cannot literally be appended to a per-trajectory representation unless converted through fixed landmarks or a declared kernel construction.

The update is

$$\mathcal{O}_{k+1}(x) = (\mathcal{O}_k(x), f_k(x)). \quad (9)$$

That one line is the architectural hinge. A legacy prototype placed synthesized pairwise functions in a registry but never appended their values to $\mathcal{O}_k$. The released engine tests the resulting matrix dimension and the appended column directly.

2.3 Declared nuisance group

For a $T \times p$ trajectory matrix X , the implemented nuisance group is

$$g \cdot X = aXQ + \mathbf{1}b^\top, \quad a > 0, \quad Q \in O(p), \quad b \in \mathbb{R}^p. \quad (10)$$

Translation, positive global scale, and orthogonal channel mixing model changes of offset, units, and sensor basis that should not change the chosen observable. Anisotropic scaling, nonlinear sensor response, time warping, resampling, and missingness are not in this group. Their treatment is empirical robustness, not exact invariance.

3 What augmentation guarantees

Theorem 3.1 (Exact fibre refinement). *Let $O_{k+1} = (O_k, f)$. Then every O_{k+1} -fibre is contained in an O_k -fibre. The refinement is strict if and only if f is nonconstant on at least one O_k -fibre.*

Proof. If $O_{k+1}(x) = O_{k+1}(x')$, equality of the first block gives $O_k(x) = O_k(x')$. Thus $\sim_{k+1}$ implies $\sim_k$. Strictness occurs exactly when some pair with $x \sim_k x'$ has $f(x) \neq f(x')$. $\square$

Theorem 3.2 (Conditional finite-sample completion). *Let a finite sample S contain n distinct trajectories, and suppose O_0 induces c_0 nonempty equivalence classes on S . If, whenever the current partition is not discrete, the hypothesis class contains a unary candidate that splits at least one nonsingleton fibre and the search returns such a candidate, then append-only synthesis makes O_k injective on S after at most $n - c_0$ successful additions.*

Proof. By [theorem 3.1](#), no current class can merge with another. Every successful candidate splits at least one class and therefore increases the number of nonempty classes by at least one. A partition of n objects has at most n classes. Hence at most $n - c_0$ increases are required. $\square$

Remark 3.3. The theorem is conditional and finite-sample. It does not prove population injectivity, global nonlinear observability, separator completeness of a practical grammar, or successful search under noise. A zero collision count on a finite audit is not an injectivity theorem.

Theorem 3.4 (Nested threshold collisions). *Fix τ and the coordinate transform h . If d_k is the max metric in [equation \(5\)](#) and $O_{k+1} = (O_k, f)$, then*

$$\{(x, x') : C_{k+1, \tau}(x, x') = 1\} \subseteq \{(x, x') : C_{k, \tau}(x, x') = 1\}. \quad (11)$$

Consequently empirical collision risk on any fixed set of distinct pairs cannot increase after an append-only update.

Proof. The new distance is

$$d_{k+1}(x, x') = \max\{d_k(x, x'), |h(f(x)) - h(f(x'))|\} \geq d_k(x, x').$$

Therefore $d_{k+1} \leq \tau$ implies $d_k \leq \tau$. The task witness D_d is unchanged. $\square$

Re-estimating a rank transform on held-out data, rescaling old coordinates after every round, changing τ , or replacing the max metric can invalidate [theorem 3.4](#). The implementation uses one fixed analytic squash precisely to avoid those loopholes.

3.1 Risk statements require independent units

All $\binom{n}{2}$ pairs from n trajectories are dependent. Treating them as independent Bernoulli trials would manufacture confidence. The outer audit therefore shuffles each binary class once and makes disjoint cross-class matches, so each trajectory contributes to at most one trial.

Proposition 3.5 (Exact upper bound after zero collisions). *For K independent Bernoulli collision trials and zero observed collisions, the exact one-sided $(1 - \alpha)$ Clopper–Pearson upper bound is*

$$p_U = 1 - \alpha^{1/K}. \quad (12)$$

At 95% confidence, $K = 120$ zero-collision trials yield $p_U = 0.0247$; approximately 299, 2,995, and 29,956 independent pairs are required to bound risk below 10^{-2} , 10^{-3} , and 10^{-4} , respectively [6]. Pairwise source selection may use the full cross-label array as a dependent U-statistic, but outer uncertainty must respect its dependence structure [15].

Proposition 3.6 (Finite-grammar audit bound). *For a fixed finite grammar $\mathcal{H}$ with $|\mathcal{H}| = M$ and K independent audit pairs, Hoeffding’s inequality and a union bound give, with probability at least $1 - \alpha$,*

$$\sup_{f \in \mathcal{H}} |\widehat{R}(f) - R(f)| \leq \sqrt{\frac{\log(2M/\alpha)}{2K}}. \quad (13)$$

This bound is intentionally conservative and applies only if the grammar and audit are genuinely fixed. It is not used to relabel the internally frozen experiment as prospective evidence.

4 Canonical structure without false completeness

4.1 Global canonicalization

For a trajectory $X \in \mathbb{R}^{T \times p}$ define

$$\tilde{X} = \frac{X - \mathbf{1}\bar{x}^\top}{\sqrt{T^{-1}\|X - \mathbf{1}\bar{x}^\top\|_F^2}}, \quad (14)$$

with the zero trajectory returned when the denominator is numerically zero. A delay cloud concatenates three copies at lags 0, 2, 4 and deterministically subsamples at most 72 points. All distance geometry is computed from this cloud.

Theorem 4.1 (Similarity equivariance of the normalized cloud). *For g in [equation \(10\)](#), nonconstant X obeys*

$$(\widetilde{g \cdot X}) = \tilde{X}Q. \quad (15)$$

Consequently all Euclidean pairwise distances in the corresponding delay cloud are unchanged.

Proof. Centering removes b . Orthogonality preserves the Frobenius norm, and $a > 0$ cancels between numerator and denominator. Delay concatenation applies the block-orthogonal action $\text{diag}(Q, Q, Q)$, which preserves pairwise Euclidean distances. $\square$

4.2 Quantile recurrence

Let d_{ij} be delay-cloud distances and let ϵ_q be their empirical q -quantile for $i < j$. With $q = 0.12$, define

$$A_{ij} = \mathbf{1}\{i \neq j, d_{ij} \leq \epsilon_q\}. \quad (16)$$

Corollary 4.2 (Quantile-recurrence invariance). *Under the nuisance group in [equation \(10\)](#), A is unchanged under a fixed deterministic quantile and tie rule. Degree summaries, transitivity, and normalized-Laplacian spectra therefore inherit the same invariance.*

The implementation checks 24 transformed trajectories and obtains identical adjacency matrices in all 24 cases. It also tests normalized-Laplacian indexing on two known graphs: $\lambda_2(P_4) = 0.5$, while disconnected $2P_2$ has $\lambda_2 = 0$. This matters because selecting the second *positive* eigenvalue silently returns λ_3 and can report nonzero “connectivity” for a disconnected graph.

4.3 Recurrence is discontinuous at the threshold

Exact invariance to the declared group is not general perturbation stability. If a perturbation changes each distance by at most η_d and moves the empirical threshold by at most η_q , only pairs satisfying

$$|d_{ij} - \epsilon_q| > \eta_d + \eta_q \quad (17)$$

are guaranteed to preserve edge status. Near-threshold edges may flip. A quantile construction also makes edge density nearly fixed, so raw density and mean degree are omitted as informative candidates; degree variation and higher-order structure remain.

4.4 Topology and temporal coordinates

Finite H_0 lifetimes are minimum-spanning-tree edge lengths, normalized by median pairwise distance. H_1 lifetimes are computed by Vietoris–Rips persistence and normalized by the same source-independent geometric scale. Persistence diagrams are stable under metric perturbation [5, 7]; stable vector summaries such as landscapes and images are established alternatives [1, 4]. Stability does not imply discriminative completeness.

The primitive library contains 18 total coordinates, of which two form the initial representation and 16 enter the grammar:

- recurrence λ_2, λ_3 , spectral entropy, degree coefficient of variation, and transitivity;
- normalized H_0 total, maximum, and entropy, plus H_1 count, maximum, and total persistence;
- radial spectral entropy, radial permutation entropy, radial turning rate, increment coefficient of variation, and covariance-spectrum entropy.

The baseline coordinates are radial lag-one correlation and path tortuosity.

Noncompleteness is explicit A logistic trajectory and its time reversal are different oriented histories, yet their recurrence normalized-Laplacian spectra agree to 3.83×10^{-15} . Graph spectra can be cospectral, persistence summaries can collide, and nuisance invariance necessarily discards information. The canonical field is a constrained representation, not a complete coordinate system.

5 The synthesis engine

5.1 A finite typed grammar

Every candidate maps one trajectory to one scalar. The depth-two grammar contains:

$$f(x) = u_i(x), \quad (18)$$

$$f(x) = \frac{u_i(x) - u_j(x)}{|u_i(x)| + |u_j(x)| + 10^{-9}}, \quad (19)$$

$$f(x) = \log \frac{|u_i(x)| + 10^{-9}}{|u_j(x)| + 10^{-9}}, \quad (20)$$

$$f(x) = \text{sgn}(u_i u_j) \sqrt{|u_i u_j|}. \quad (21)$$

Primitive complexity is one; a binary expression has complexity three. Geometric means are restricted to adjacent primitives in deterministic sorted order. The resulting manifest contains 271 programs. Protected operations return finite values and clip only beyond magnitude 10^6 .

This is symbolic feature construction, but its objective differs from ordinary equation fitting. Symbolic regression searches for compact relations in data [25, 29]; SINDy searches sparse governing equations [2]. Here the output is an observation coordinate chosen to split counterexample fibres while surviving a nuisance gate.

5.2 Selection rule

Within each source domain, records are stratified into 60% selection and 40% audit subsets. A candidate must satisfy, across every source domain:

$$e_{\text{inv}}(f) \leq 5 \times 10^{-6}, \quad (22)$$

$$Q_{0.95}[e_{\text{noise}}(f)] \leq 0.20, \quad (23)$$

$$\min_d \Delta R_d^{\text{audit}}(f) \geq 0.15, \quad (24)$$

$$\min_d \text{AUC}_{\pm}^{\text{audit}}(f) \geq 0.75, \quad (25)$$

$$\min_d \text{AUC}_{\pm}^{\text{select}}(f) \geq 0.70, \quad (26)$$

where orientation-free AUC is $\max(\text{AUC}, 1 - \text{AUC})$. Surviving candidates are ordered deterministically by

$$\begin{aligned} U(f) = & \min_d \Delta R_d^{\text{audit}} + 0.50 \text{ mean}_d \Delta R_d^{\text{audit}} + 0.15 \min_d \Delta R_d^{\text{select}} \\ & + 0.05(\text{mean}_d \text{AUC}_{\pm}^{\text{audit}} - 0.5) - 0.003 \text{ complexity}(f) - 0.10 Q_{0.95}[e_{\text{noise}}(f)]. \end{aligned} \quad (27)$$

The winner is appended, never substituted for an old coordinate. Search stops when no candidate passes or after three additions. In the released runs it stops after one.

5.3 Executable loop

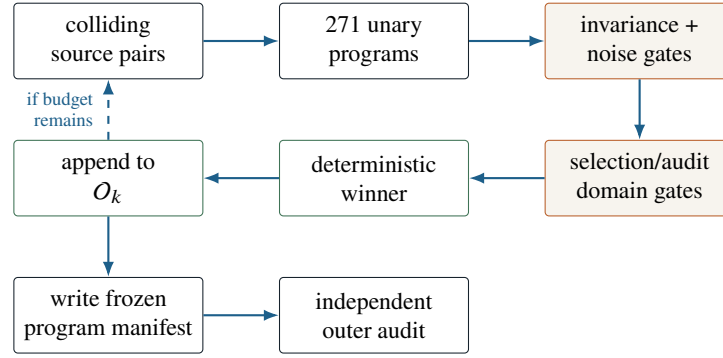


Figure 1: The pressure loop. Source pairs drive candidate choice; the program, preprocessing, threshold, and split declaration are serialized before independent synthetic audit and real-test loading within the final run. The freeze has no external timestamp.

The selected all-source program is

$$f_{\star}(X) = \log \frac{\text{CV}(\|\Delta\tilde{X}_t\|) + 10^{-9}}{H_{\text{perm}}(\|\tilde{X}_t\|) + 10^{-9}}, \quad (28)$$

the log ratio of increment coefficient of variation to radial permutation entropy. Its maximum relative discrepancy under the declared similarity transform is 2.58×10^{-14} over every trajectory in the final suite.

5.4 Why the replacement engine was necessary

The historical prototype contributed the conceptual loop but not valid evidence. Forensic recovery showed that its synthesized feature was not reinserted; it mixed unary and pairwise programs; indexed recurrence connectivity incorrectly; used one perturbation so a variance-based stability score was identically one; and allowed nominal transfer domains into optimization. Its retained seismic pilot additionally lacks raw data and contains target-selection and normalization leakage. All legacy quantitative claims are excluded. The release preserves the exact commit and defects in a dedicated audit so that the development path is inspectable rather than erased.

6 Evidence contract

6.1 Controlled discovery and independent audit

The discovery suite has three qualitatively different generators, using canonical logistic, Lorenz, and coupled-oscillator models as controlled test beds [17, 18, 21]:

- **Logistic:** $r \in [3.20, 3.42]$ versus $r \in [3.90, 3.99]$, 256 retained scalar samples after burn-in, Gaussian noise $\sigma = 0.003$.
- **Lorenz:** $\rho \in [17, 21]$ versus $\rho \in [27, 31]$, RK4 at $\Delta t = 0.008$, 256 retained three-channel samples, Gaussian noise $\sigma = 0.01$.
- **Kuramoto:** five oscillators with coupling $K \in [2.8, 4.0]$ versus $K \in [0.05, 0.45]$, 256 retained sine-channel samples, Gaussian noise $\sigma = 0.015$.

Discovery uses 28 trajectories per class per system. For each outer fold, two systems are sources. The held-out system is regenerated after program freeze with a separate final seed and 120 trajectories per class. One random matching creates 120 disjoint cross-label pairs. No audit trajectory participates in source feature scaling, grammar construction, candidate gates, or stopping.

The two baseline features, fixed squash, $\tau = 0.22$, group action, 271-program grammar, three-round budget, and gate constants are machine-readable. Source selection uses all cross-label pairs only inside source data. The outer statistic uses independent matched pairs and a paired 4,000-repetition bootstrap on before/after indicators.

6.2 Nulls and ablations

Three different controls answer different questions:

1. **Fixed-program label permutation:** permute audit labels 999 times while holding the chosen program fixed.
2. **Full-pipeline source-label null:** permute source labels, rerun all synthesis gates, then apply any selected program to the untouched independent audit. Ninety-nine repetitions set a minimum attainable p -value of 0.01 using the plus-one rule [22].
3. **Admissible random-program comparator:** draw one program that passes source invariance and noise gates, apply it directly to the outer audit, and repeat 999 times. This is not a rerun of source-domain selection and is not described as equal-compute.

Primitive-only, recurrence-only, and topology-only grammars are rerun through the same one-round source selection. Phase-randomized and IAAFT surrogates [26, 28], time warps, irregular sampling, and full learned-representation baselines remain specified extensions rather than completed evidence.

6.3 Real-data stress, sealed only within the run

After the all-source program is serialized, the runner reads only the official UCR archive TEST members for ECGFiveDays and Earthquakes [9]. It deterministically samples up to 32 cases per class, producing 64 trajectories and 32 disjoint cross-label pairs per dataset. Archive SHA-256 digests, exact selected identifiers, and acquisition URLs are shipped. No real record changes the expression, constants, threshold, or preprocessing.

This is a stringent zero-adaptation stress test but not a prospective registered study. The task meanings also differ: ECG morphology and a constructed earthquake-event classification problem are not draws from a well-defined population of “all dynamical systems.” Results are therefore reported by name, not generalized to an imaginary superpopulation.

7 Results

7.1 Independent synthetic audits

Table 2: Independent synthetic audit. Each row uses 120 disjoint cross-label pairs generated with a seed not used in discovery. Intervals are paired bootstrap 95% intervals for absolute reduction.

Held out	Before	After	ΔR	95% interval	AUC	Selected program
Logistic	0.992	0.000	0.992	[0.975, 1.000]	1.000	log(increment CV/radial PE)
Lorenz	0.917	0.467	0.450	[0.358, 0.533]	0.828	log(covariance entropy/ H_1 max)
Kuramoto	1.000	0.083	0.917	[0.867, 0.967]	0.998	log(increment CV/radial PE)
Macro mean	0.969	0.183	0.786	—	0.942	fold-specific frozen programs

The logistic fold has zero post-repair collisions, giving a one-sided exact 95% upper risk bound of 0.0247, not zero population risk. Lorenz retains 56/120 collisions and is the clearest controlled failure mode. Kuramoto retains 10/120. All three fixed-program permutation values are below 0.05 (0.001, 0.027, and 0.001 for logistic, Lorenz, and Kuramoto). More importantly, the full source-label pipeline selects no program in 0/99 permutations for each fold; the plus-one p -value is 0.01.

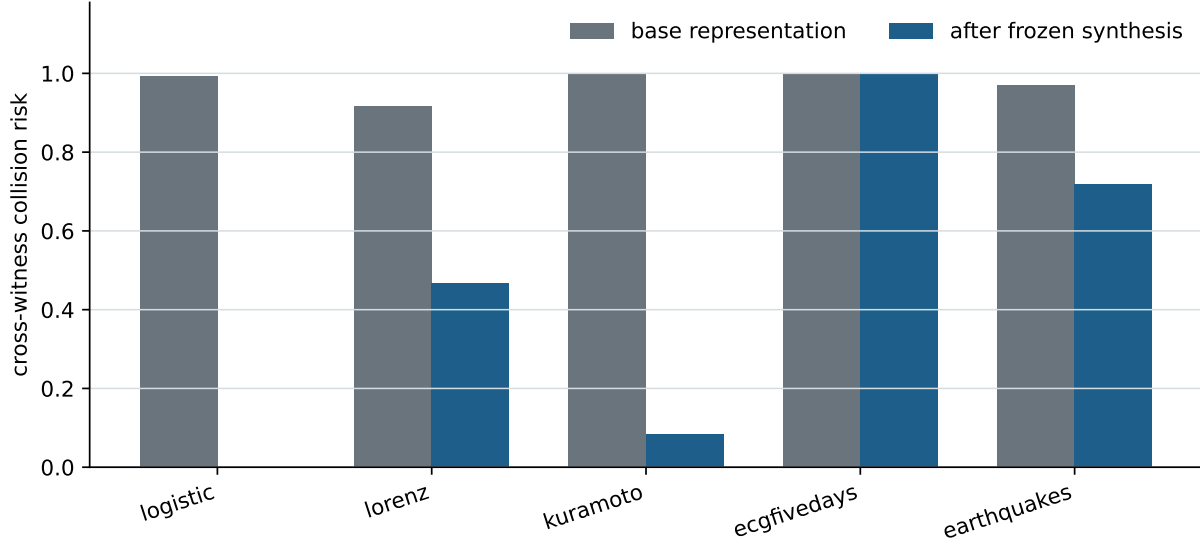


Figure 2: Cross-witness collision risk before and after the frozen fold-specific program for controlled synthetic audits (first three groups), followed by the all-source program on two real UCR stress tests. The visual discontinuity is substantive: strong controlled-domain repair does not imply real-domain transfer.

7.2 The random-program comparator and ablations

The selected program exceeds the admissible random-program distribution significantly only for Kuramoto: comparator p -values are 0.119 for logistic, 0.166 for Lorenz, and 0.034 for Kuramoto. Therefore the data support successful gated synthesis, but not systematic superiority over a randomly drawn admissible grammar member across all three domains.

Table 3: Absolute collision-risk reduction under one-round grammar ablations on the independent synthetic audits. A zero may mean no program passed the source gates.

Held out	Full grammar	Primitive only	Recurrence only	Topology only
Logistic	0.992	0.000	0.133	0.000
Lorenz	0.450	0.133	0.000	0.383
Kuramoto	0.917	0.000	0.000	0.292

No single family dominates every system. Under these source-selection gates, the symbolic composition is essential to transfer into logistic and Kuramoto, while a topology-only expression approaches the full Lorenz reduction. This supports a heterogeneous primitive library; it does not establish that these are optimal primitives or that the selected composition is the only target-domain separator.

7.3 A post hoc mechanistic read of the Kuramoto fold

COMPUTED; POST HOC. Kuramoto is the only fold in which the source-selected expression significantly exceeds the admissible random-program comparator. The winner was chosen from logistic and Lorenz records and frozen before any Kuramoto audit trajectory was generated. After that result was known, the exact 240-record audit was regenerated bit-for-bit—the maximum observed-channel discrepancy is zero—while retaining latent phases that were never exposed to synthesis.

For the conventional generalized order parameters and their time averages, define [8]

$$Z_m(t) = \frac{1}{N} \sum_{j=1}^N e^{im\theta_j(t)}, \quad \bar{R}_m = \frac{1}{T} \sum_t |Z_m(t)|. \quad (29)$$

The noiseless sine sensor vector $s_j(t) = \sin \theta_j(t)$ obeys the exact identity

$$\|s(t)\|_2^2 = \frac{N}{2} [1 - \Re Z_2(t)]. \quad (30)$$

The released observed array adds Gaussian measurement noise after this sine transform, so [equation \(30\)](#) is not an exact identity for the array supplied to synthesis. Even without that noise, channel centering adds the explicit correction $-2 \sum_j \bar{s}_j \sin \theta_j(t) + \sum_j \bar{s}_j^2$, after which the declared global RMS normalization is applied. The identity therefore supplies a mechanistic route by which radial permutation entropy can remain sensitive to generalized phase-order structure, while increment CV summarizes temporal heterogeneity of the multichannel motion. It does not identify the computed feature with Z_1 or Z_2 .

On the exact audit, the frozen scalar has orientation-free label AUC 0.9976; latent $\bar{R}_1$ and $\bar{R}_2$ each have AUC 0.9989. The scalar's pooled Spearman correlation is 0.861 with $\bar{R}_1$ (stratified record-bootstrap 95% interval [0.833, 0.885]) and 0.907 with $\bar{R}_2$ (interval [0.885, 0.927]). The second-harmonic association sharpens within the weak-coupling class: $\rho_S = 0.944$ (record-bootstrap interval [0.908, 0.963]), versus 0.888 for increment CV and -0.864 for radial permutation entropy. In a paired record bootstrap, the composite-minus-increment correlation difference is 0.056 ([0.029, 0.095]), and the composite-minus-oriented-entropy difference is 0.080 ([0.043, 0.128]). These intervals were designed after inspection and are descriptive, not confirmatory.

This is a narrow within-regime gain, not evidence that symbolic composition is required for the coarse target contrast. Across the two widely separated coupling classes, increment CV alone has AUC 0.9992, slightly above the composite. Its primitive-only ablation failed the *source* gates; it was not intrinsically uninformative on Kuramoto. The multichannel relation is nevertheless essential to the released scalar: after recomputing canonicalization and both primitives for every one of the $\binom{5}{m}$ channel subsets, mean label AUC rises from 0.522 for one channel to 0.950, 0.992, 0.998, and 0.998 for two through five channels.

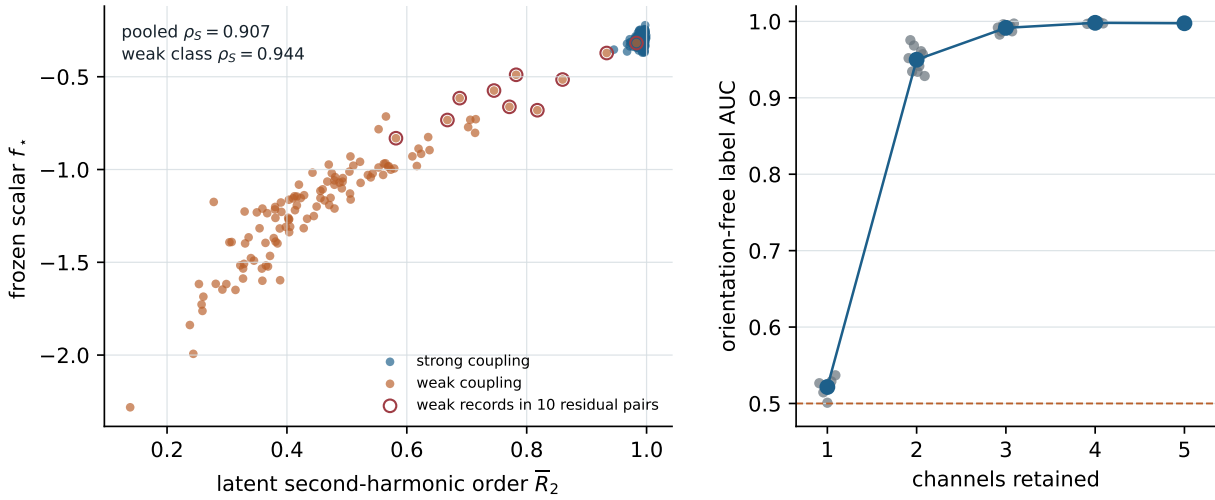


Figure 3: Post-hoc interpretation of the frozen Kuramoto winner. Left: the source-selected scalar tracks latent second-harmonic coherence $\bar{R}_2$; rings mark the weak-coupling records in the ten residual released-match collisions. Right: every point is one of the $\binom{5}{m}$ channel subsets, with canonicalization and both primitives recomputed for that subset; blue points are subset-count means. These are descriptive diagnostics on a result already seen, not a new held-out test.

The ten weak-coupling records in the residual released matching are unusually ordered within their declared class: mean $\bar{R}_1$ is 0.874, versus 0.570 for the other weak-coupling records, while mean K is 0.346 versus 0.237. The conclusion is not an accident of one matching: all 14,400 cross-label pairs have post-repair risk 0.091, and 20,000 random perfect matchings give a descriptive 95% risk range [0.075, 0.108]; the released 10/120 = 0.083 lies inside it.

The bounded interpretation is therefore that a scalar selected without Kuramoto access transferred as an order-sensitive, sensor-basis-invariant surrogate inside this finite controlled generator. Its strongest post-hoc association is with generalized second-harmonic coherence, and the ratio combines complementary temporal signals for within-weak-class ranking. The analysis is post hoc, $N = 5$, the coupling intervals are widely separated, the expression's orientation reverses across other systems, and a simpler target-domain primitive separates the coarse classes at least as well. It is not a conserved invariant, a critical-coupling estimate, a universal order coordinate, or evidence of information beyond conventional order summaries.

7.4 Invariance and robustness audits

The all-source expression in [equation \(28\)](#) passes its source selection noise gate with 95th-percentile relative error $0.110 < 0.20$. Across synthetic audits and both real sets, however, the same statistic has median error 0.034, 95th percentile 0.265, and

maximum 0.415 under 2% relative Gaussian perturbations. The source gate therefore does not guarantee unseen-domain robustness. Exact group invariance remains intact; off-group noise robustness is empirical and can fail.

The recurrence adjacency check is exact in 24/24 similarity transforms, the selected expression’s maximum relative transform error is 2.58×10^{-14} , and the graph-spectrum unit cases return the correct λ_2 . All ten core tests pass.

7.5 Real stress tests do not confirm transfer

Table 4: All-source expression on UCR TEST records loaded after the in-run freeze. Each row has 32 disjoint pairs. The permutation test holds the expression fixed and permutes labels 999 times.

Dataset	Before	After	ΔR	95% interval	AUC	Permutation p
ECGFiveDays	1.000	1.000	0.000	[0.000, 0.000]	0.502	1.000
Earthquakes	0.969	0.719	0.250	[0.125, 0.406]	0.731	0.410

The Earthquakes point estimate is positive but ordinary under the fixed-program permutation distribution ($p = 0.41$). ECG-FiveDays is null. These results close the strongest version of the release claim: the discovered expression is not demonstrated as a universal cross-domain observable. The real track succeeds as a falsification mechanism because it prevents controlled synthetic performance from being narrated as broad physical discovery.

8 What is new, and what is borrowed

Recurrence plots and recurrence networks convert trajectory geometry into discrete structure [11–13, 20]. Persistent homology summarizes multiscale topology. DMD and its extended variants approximate Koopman structure [24, 30], while HAVOK uses delay coordinates to isolate intermittent forcing [3]. Compact time-series feature sets and modern learned encoders provide powerful alternative representations [10, 19, 31]. Counterexample-guided inductive synthesis alternates candidate construction and verification [16].

No individual component above is claimed as new. The contribution is their organization around a specific representational failure:

$$\boxed{\text{task-distinct collision} \rightarrow \text{typed unary search} \rightarrow \text{nuisance gate} \rightarrow \text{actual reinsertion} \rightarrow \text{sealed retest.}} \quad (31)$$

This changes the scientific object. The aim is not simply accurate classification, equation recovery, or a complete invariant. It is an auditable repair to the observation map that responds to a witnessed failure and carries a proof of fibre refinement.

The strongest novelty claim is therefore architectural and methodological: collision counterexamples, nuisance constraints, interpretable symbolic programs, append-only monotonicity, and domain holdout are joined in one executable loop. A stronger claim—that this loop reliably outperforms generic search or discovers physical laws—is open.

9 Failure modes and open work

1. **Task dependence.** D_d supplies supervision. The method constructs predictive or regime-relevant observables; it does not discover relevance without a witness.
2. **Invariant is group-relative.** The exact theorem covers translation, positive global scale, and orthogonal channel action only. It does not cover anisotropic units, nonlinear sensing, arbitrary resampling, or time warping.
3. **Canonical is not complete.** Time reversal and cospectral graphs provide explicit collisions. Persistent summaries also compress.
4. **Finite search is conditional.** The completion theorem assumes a separator exists and is returned. The 271-program grammar is deliberately small.
5. **Threshold dependence.** $\tau = 0.22$ is fixed for this audit. A threshold curve and source-only calibration study should precede future prospective use.
6. **Internal freeze only.** The final runner serializes the program before audit loading, but the project development used an earlier null failure to improve gates. The final seeds are new; the study is still not externally timestamped or independently replicated.
7. **Limited permutation resolution.** The full-pipeline null has 99 repetitions. A confirmatory run should use at least 9,999 and a predeclared primary statistic.

8. **Comparator scope.** Random-program comparison samples admissible expressions directly; it is not a compute-matched alternative optimizer. Standard symbolic regression, sparse selection, catch22, ROCKET, TS2Vec, DMD/EDMD/HAVOK, and DTW remain future baselines.
9. **Synthetic independence is not physical diversity.** Three generators do not define a population of physical domains. Their clear regimes may reward low-complexity statistics.
10. **Real transfer is unconfirmed.** Neither real stress test is significant. More datasets should be grouped by genuinely independent units such as subject, event, device, or acquisition session.
11. **No law discovery.** The selected expression is not shown to be conserved, causal, or mechanistically fundamental. Known first-integral tracks with oracle invariants are required before using that phrase.
12. **Legacy seismic work is exploratory only.** Missing raw data and confirmed leakage preclude its use as evidence here.

The next decisive experiment is prospective leave-one-domain-out evaluation with an external timestamp, a frozen threshold curve, event/subject grouping, 9,999 full-pipeline permutations, and equal-budget baselines. Positive transfer should be replicated on a second domain family before any general claim.

10 Conclusion

Paper I identifies the obligation to preserve distinctions through dynamics and observation. Paper II asks whether a distinction hidden by the current representation changes what can be predicted. Paper III supplies the constructive response: let the failed fibre pressure the representation itself to change.

The formal contribution is an append-only refinement calculus. Exact fibres refine, max-metric collision sets nest, and a separator-complete search terminates on a finite sample. The representational contribution is a global trajectory canonicalization whose recurrence geometry is exactly invariant to a declared similarity group, paired with an explicit proof that the resulting summaries are not complete. The computational contribution is an engine in which a typed unary program is selected under invariance, noise, source-audit, discrimination, and complexity gates and then genuinely reinserted.

On independent synthetic audits, that mechanism reduces macro collision risk by 0.786, with full-pipeline source-label nulls selecting nothing. A post-hoc dissection shows that the strongest search-specific result, Kuramoto, transfers an order-sensitive multichannel surrogate, while also showing that the composite is not unique on the target records. The Lorenz residual risk, random-program results, off-group noise degradation, and null real transfer prevent an inflated conclusion. What has been shown is not universal invariant discovery. It is something narrower and, because it is falsifiable, more durable: a representation can be repaired by its own meaningful collisions, the repair can be made interpretable and group-aware, and the boundary between controlled success and unconfirmed transfer can be exposed by design.

A Additional proof details

A.1 Quantile recurrence before canonicalization

Suppose a finite point cloud is transformed as $y_i = aQx_i + b$ with $a > 0$ and Q orthogonal. Then

$$\|y_i - y_j\|_2 = a\|x_i - x_j\|_2.$$

Every order statistic and empirical quantile of the distance multiset also scales by a . Thus

$$\|y_i - y_j\|_2 \leq a\epsilon_q \iff \|x_i - x_j\|_2 \leq \epsilon_q.$$

Global canonicalization removes a first, making thresholds equal as well. At exact repeated distances, invariance requires the same deterministic weak-inequality and quantile interpolation convention. Floating-point reductions may differ at machine precision; the direct adjacency audit tests the actual implementation.

A.2 Persistence scale and similarity action

Vietoris–Rips filtrations depend only on pairwise distances. Translation and orthogonal action leave birth and death scales unchanged; positive scaling multiplies them by a . Dividing lifetimes by a nonzero distance scale that also multiplies by a gives a scale-invariant summary. In the implementation the cloud is already globally normalized, and the remaining division by median pairwise distance makes the convention explicit. Stability under small perturbations follows at the diagram level under standard bottleneck/Gromov–Hausdorff hypotheses; hard lifetime thresholding at 0.03 can still change a feature near that cutoff.

A.3 Why recurrence spectra ignore time direction

Let P reverse the order of delay-cloud vertices. Ignoring boundary differences already fixed by constructing the reversed cloud, recurrence adjacency transforms by $A' = PAP^T$. The normalized Laplacian obeys $L' = PLP^T$ and has the same eigenvalues. Thus an orientation-sensitive history can be lost even when every eigenvalue is retained. Temporal primitives are included partly to answer this known blind spot; they do not guarantee completeness.

B Primitive definitions and implementation contract

Table 5: Implemented primitive families. “Invariant” means under [equation \(10\)](#), aside from floating-point and threshold-tie handling.

Family	Coordinates	Contract
Baseline	radial lag-one correlation; path tortuosity	Always retained; not searched away.
Recurrence network	λ_2, λ_3 , spectral entropy, degree CV, transitivity	Quantile $q = 0.12$; at most 72 delay vertices; correct zero multiplicity.
Persistent topology	H_0 total/max/entropy; H_1 count/max/total	Rips H_1 ; finite lifetimes; median-distance normalization; lifetime cutoff 0.03.
Temporal radial	spectral entropy, permutation entropy, turning rate	Operate on $\ \tilde{X}_t\ $; order sensitive except spectral magnitude symmetries.
Increment	coefficient of variation	Operates on $\ \tilde{X}_{t+1} - \tilde{X}_t\ $.
Covariance	eigenvalue entropy	Orthogonal-basis invariant; dimension can influence interpretation.

The supported software contract is Python 3.11 or later with locked numerical dependencies; the canonical release run used Python 3.14.6. Feature extraction is deterministic. The evidence runner fails if either raw archive is absent, writes a grammar manifest and declared split manifest, freezes selected programs, then materializes the independent audit and real TEST records. Ten unit tests cover recurrence invariance, λ_2 , reinsertion, threshold nesting, finite-sample splitting, the exact zero-collision bound, deterministic generation, TEST-only archive loading, exact Kuramoto regeneration, and the sine/order identity.

C Result ledger

Table 6: Compact result ledger. Paths are relative to the Paper III directory.

Status	Claim	Primary artifact
PROVED	Append-only augmentation refines fibres; successful finite synthesis terminates under separator completeness; max-metric collisions nest.	Manuscript section 3 ; <code>test_pdis_core.py</code> .
PROVED	Global canonicalization and quantile recurrence are invariant under the declared similarity group.	Manuscript section 4 ; <code>canonical.py</code> .
COMPUTED	24/24 adjacency checks; correct graph λ_2 ; selected-program transform error 2.58×10^{-14} .	<code>evidence/outputs/metrics.json</code> .
HELD-OUT	Synthetic reductions 0.992, 0.450, 0.917 on 120 disjoint pairs per domain.	<code>transfer_metrics.csv</code> ; frozen and split manifests.
COMPUTED	Full-pipeline source-label null selects a program in 0/99 repetitions per fold.	<code>metrics.json</code> ; <code>DEVELOPMENT_AUDIT.md</code> .
COMPUTED; POST HOC	The frozen Kuramoto scalar is an order-sensitive multichannel surrogate on the exact audit; no claim upgrade.	<code>kuramoto_posthoc.json</code> ; section 7.3 .
HELD-OUT	ECG reduction 0, Earthquakes reduction 0.25 with $p = 0.41$; transfer unconfirmed.	<code>metrics.json</code> ; raw archive hashes.
OPEN	Generalization to unseen physical domains, superiority to competitive optimizers, and recovery of physical laws.	Future externally committed study.

D Reproduction

From the Paper III directory, the complete audit is:

```
uv sync --frozen
uv run pytest -q
uv run python evidence/run_all.py
shasum -a 256 -c evidence/outputs/manifest.sha256
```

The runner regenerates JSON, CSV, and all three paper figures. PDF creation metadata can vary across plotting backends; the manifest is a release-integrity record for the committed artifacts, not an external commitment. Raw data licenses and source terms remain those of the UCR archive and original contributors.

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